

REVIEW OPEN ACCESS

Mid-Infrared Integrated Photonics: Material Platforms and Emerging Applications

Muhammad Ali Butt¹  | Daniel Pastor Abellán² | Ryszard Piramidowicz¹

¹Institute of Microelectronics and Optoelectronics, Warsaw University of Technology, Koszykowa 75, Warsaw, Poland | ²Photonic IC-group at the Photonics Research Labs, Universitat Politècnica de València, C/ Camino de Vera s/n, Valencia, Spain

Correspondence: Muhammad Ali Butt (ali.butt@pw.edu.pl)

Received: 15 January 2025 | **Revised:** 21 October 2025 | **Accepted:** 22 October 2025

Keywords: mid-infrared | integrated photonics | material platforms | spectroscopy

ABSTRACT

Mid-infrared (MIR) integrated photonics plays a pivotal role in a range of scientific and technological applications due to the unique absorption features of many molecules within this spectral range (2–20 μm). These wavelengths are particularly useful for sensitive chemical and biological sensing, as well as in environmental surveillance, medical diagnostics, and industrial process control. Advances in material platforms such as chalcogenide glasses, silicon, germanium, and graphene have significantly enhanced the performance and miniaturization of MIR integrated photonic devices. In this review, we provide a comprehensive analysis of various material platforms utilized in MIR integrated photonics. Additionally, the review explores the broad range of potential applications of MIR integrated photonics, including but not limited to environmental sensing, molecular spectroscopy, free-space communication, and thermal imaging. The capabilities of MIR integrated photonic devices for precise chemical detection, enhanced biomedical diagnostics, and industrial process supervising are highlighted, emphasizing the transformative impact of these technologies on fields such as environmental science, healthcare, and telecommunications.

1 | Introduction

Mid-infrared (MIR) photonics is a rapidly evolving field with profound significance in a variety of scientific, industrial, and medical applications [1–4]. Operating in the 2 to 20 μm wavelength range, MIR light interacts strongly with molecular vibrations, making it ideal for spectroscopy and chemical sensing [5, 6]. One of the key advantages of MIR photonics is its ability to detect and identify specific molecular compounds based on their unique absorption spectra, which makes it indispensable for gas sensing, environmental monitoring, and industrial process control [7–10]. In healthcare, MIR techniques are used for non-invasive diagnostics, such as detecting glucose levels or early-

stage cancer biomarkers by analyzing breath or skin samples [11, 12]. Moreover, MIR photonics is critical in defence and security, enabling the development of thermal imaging systems for night vision, target identification, and missile tracking. Advances in MIR lasers and detectors have further broadened its applications, facilitating research in astrophysics, where MIR wavelengths allow astronomers to study celestial objects obscured by cosmic dust. In telecommunications, MIR photonics is pushing boundaries, offering new avenues for high-speed data transmission over longer distances due to the lower scattering losses of longer wavelengths [13–17]. Additionally, MIR light is valuable in material processing, including laser surgery and precision cutting in manufacturing, due to its ability to interact strongly

Abbreviations: CaF_2 , calcium fluoride; CMOS, complementary metal-oxide-semiconductor; Fs, Femtosecond; FTIR, fourier transform infrared; IR, infrared; IL, insertion loss; ICL, interband cascade laser; LEs, interlayer excitons; LN, lithium niobate; MIR, mid-infrared; PD, photodetector; PIC, photonic integrated circuit; PMMA, poly (methyl methacrylate); QCL, quantum cascaded laser; Si, silicon; SiC, silicon carbide; SiN, silicon nitride; SOI, silicon-on-insulator.

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2025 The Author(s). *Laser & Photonics Reviews* published by Wiley-VCH GmbH

with certain materials without extensive thermal damage [18, 19]. As photonic devices and materials, such as quantum cascade lasers (QCLs) [20, 21], interband cascade lasers (ICLs) [22], and nonlinear crystals [23], continue to advance, the potential of MIR photonics is expanding, driving innovation across sectors and addressing critical technological challenges in sensing, imaging, and communications [24–31]. The main characteristics of MIR photonics are presented in Table 1.

In the MIR wavelength spectrum, suitable material platforms for integrated optics must exhibit transparency and low optical losses at these wavelengths, as well as compatibility with fabrication techniques for photonic devices, as discussed in the next section. Silicon nitride PICs have garnered significant attention for their exceptional ability to minimize propagation loss while offering superior potential for integration [45]. Silicon-on-insulator (SOI) is commonly used for MIR applications up to around 8 μm due to its favorable refractive index and well-established processing methods [5]. Beyond this range, Si begins to absorb, and alternative materials like Ge, which is transparent up to 14 μm [46], and chalcogenide glasses, which offer broad transparency from 2 to 12 μm , are utilized [47]. III-V semiconductors, such as gallium arsenide (GaAs) [48] and indium phosphide (InP) [49], are also promising due to their high refractive indices and capability to integrate active components like lasers and detectors. Additionally, sapphire and calcium fluoride (CaF_2) are used for specific MIR applications due to their wide transparency windows and chemical robustness [50]. The choice of platform depends on the specific wavelength range, required integration level, and the type of photonic components being evolved. The optical and physical characteristics of several material platforms are discussed in Table 2. The structure of this paper is outlined as follows:

Section 2 delves into a comprehensive discussion of the various material platforms utilized in MIR photonics. This section examines a wide array of platforms, including silicon-based technologies, silicon nitride, and silicon carbide, as well as germanium and germanium-on-silicon. It also covers chalcogenide glasses, III-V semiconductors, emerging 2D materials, and calcium fluoride, alongside other crystalline materials that play a crucial role in MIR photonic applications. In Section 3, the focus shifts toward the diverse and growing applications of MIR photonics. These applications span several critical domains such as chemical and environmental sensing and biomedical technologies. Additionally, the section highlights MIR's role in optical communications, agricultural technologies, and the food industry, along with its importance in laser-based manufacturing and energy-related solutions. Section 4 presents an in-depth discussion of the key challenges facing the development and widespread adoption of MIR photonics, while also exploring the prospects and technological advancements needed to overcome these hurdles. Finally, the paper concludes in Section 5, summarizing the key insights and offering final reflections on the state of MIR photonics and its potential trajectory.

2 | Material Platforms for MIR Photonics

The development of photonic integrated circuits (PICs) in the MIR spectral range is dependent on the availability of suitable

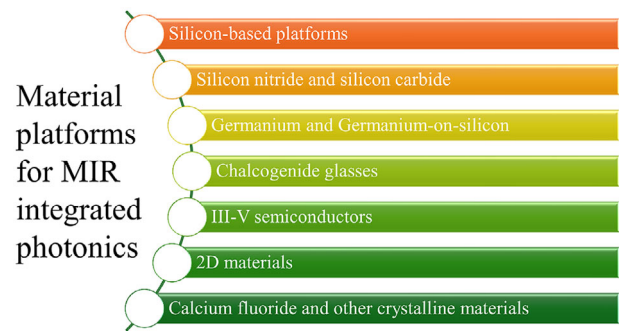


FIGURE 1 | Material platforms for MIR integrated photonics.

material platforms that support low-loss light propagation, efficient light generation, modulation, and detection [33]. The choice of materials for MIR-integrated photonics is critical, as it dictates the device performance and determines the wavelength coverage. Several material platforms have been explored to support MIR photonics, each with unique advantages and limitations [33]. Figure 1 presents the main material platforms that are discussed in this paper.

2.1 | Si-Based Platforms

Si photonics has emerged as a promising platform for the next generation of integrated photonic devices, driven by its low optical loss, mature fabrication processes, and the inherently high RI of Si [69]. Historically, the majority of Si photonics research has focused on telecom wavelengths (NIR) [70, 71], where substantial advancements have been made, particularly in optical interconnects and sensing applications. Soref's seminal work extended this paradigm by suggesting that Si is not only well-suited for NIR applications but also for the MIR spectrum, owing to Si's broad optical transparency window ranging from 1.1 to 8 μm [72].

While the field of NIR Si photonics is well-established, research into MIR Si photonics is still in its nascent stages. However, the high RI of Si continues to be a key enabler for the dense integration of photonic components, and its compatibility with CMOS processes allows for scalable, cost-effective manufacturing. Moreover, in the MIR regime, the impact of two-photon absorption—a limiting factor in NIR becomes negligible, unlocking the potential for nonlinear optical effects, such as broadband light source generation. These factors collectively position MIR Si photonics as a highly attractive area for future exploration.

SOI technology has been widely adopted for NIR applications due to its ease of fabrication and performance [73]. However, in the MIR range, SOI platforms encounter significant challenges, particularly due to strong absorption losses in the buried oxide layer between 2.6–2.9 μm and beyond 3.6 μm . To overcome these limitations and fully harness Si's transparent window, alternative designs such as free-standing structures have been proposed. These include suspended-membrane Si rib WGs [74, 75] and subwavelength grating (SWG) WGs [76, 77], both of which have been experimentally demonstrated to extend the operational wavelength up to 8 μm . However, these approaches come with trade-offs, including increased fabrication complexity, structural

TABLE 1 | Characteristics of MIR photonics.

Characteristics	Description
Spectral range	MIR photonics operates in the 2 μm to 20 μm wavelength range [32].
Applications	Used in spectroscopy, environmental monitoring, chemical sensing, biomedical imaging, security, and free-space communication [1, 33].
Key Materials	Semiconductor materials like InSb, InAs, InGaAs, HgCdTe; chalcogenide glasses, silicon, and germanium are common for MIR devices [24, 33].
Photon Sources	Lasers such as QCLs, ICLs, LEDs, and thermal emitters are used as MIR photon sources [32, 34, 35].
Photon Detectors	Common detectors include HgCdTe (MCT), InSb, PbSe, and thermoelectric detectors, such as microbolometers.
Advantages	MIR wavelengths are strongly absorbed by molecular vibrations, making them ideal for sensing specific gases and chemical compounds with high sensitivity [2, 3, 10].
Challenges	High material absorption and loss, lack of cost-effective, efficient, room-temperature devices, and complex fabrication processes increase the cost per device, limiting commercial adoption, especially in emerging markets [36].
Atmospheric Transparency Windows	3–5 μm and 8–12 μm atmospheric transmission windows are important for free-space communication and remote sensing applications [37, 38].
Nonlinear Optical Effects	Strong nonlinear effects due to longer wavelengths enable efficient frequency conversion, parametric generation, and sensing techniques [23].
Spectroscopic Significance	MIR light interacts with the vibrational modes of molecules, allowing for detailed molecular fingerprinting for gas sensing, pollution monitoring, and healthcare diagnostics [39–41].
Thermal Imaging	MIR wavelengths are used for thermal imaging because many objects at ambient temperatures emit thermal radiation in this range [42].
Energy Efficiency	QCLs in the MIR region are efficient and tunable, making them important for industrial sensing and spectroscopy applications [43, 44].

TABLE 2 | Optical and physical characteristics of several material platforms reported in this paper.

Platforms	Transparency Range (μm)	RI	Thermal Properties	Fabrication Compatibility	Key Applications	Prominent advantages	Limitations
Silicon (Si)	1.1–8 [51]	~ 3.4	High thermal conductivity ($\sim 140 \text{ W/m}\cdot\text{K}$) [52]	complementary metal-oxide-semiconductor (CMOS) compatible	Waveguides (WGs), modulators, sensors [3, 53]	Mature technology, high-index contrast	Absorbs beyond $8 \mu\text{m}$
Silicon Nitride (SiN)	0.4–7 [54]	~ 2.0	High thermal stability, lower conductivity	CMOS compatible [55]	Integrated photonics, nonlinear optics [56]	Low propagation loss, transparent to MIR	Fabrication stress, lower RI
Silicon Carbide (SiC)	0.4–17+ [57]	~ 2.6	High thermal conductivity ($\sim 120 \text{ W/m}\cdot\text{K}$)	CMOS and MEMS compatible [58]	Power electronics, photonics, THz applications, biosensors [57]	Wide transparency, high thermal durability	Higher fabrication cost
Germanium (Ge)	1.8–14 [59]	~ 4.0	Moderate thermal conductivity ($\sim 43 \text{ W/m}\cdot\text{K}$) [52]	Some CMOS compatibility	Photodetectors (PDs), WGs, IR optics	High RI, MIR transparency	Oxidation sensitivity, challenging fabrication
Germanium-on-Silicon (Ge-on-Si)	2–14 [60]	~ 3.4 (Ge on Si)	Improved thermal handling	Integration with Si photonics	Detectors, modulators, integrated circuits	Si platform integration, improved performance	Strain and defect management needed
Chalcogenide Glasses	3–5 and 8–12 [61]	2.0–3.2	Low thermal conductivity	Non-CMOS compatible	IR sensors, optical fibers, nonlinear optics	Broad MIR transparency, high nonlinearity	Low thermal stability, difficult to integrate
III-V Semi-conductors	1–10+ [62]	2.8–3.5 (varies)	High thermal conductivity, tunable	Limited CMOS compatibility	Lasers, detectors, modulators, PICs	Tunable bandgap, high-efficiency	Expensive, complex fabrication processes
2D Materials (e.g., graphene)	0.3–20+ [63]	Tunable (~ 2.5 –4.0)	High thermal conductivity (graphene $\sim 5000 \text{ W/m}\cdot\text{K}$) [64]	Limited CMOS compatibility	Modulators, detectors, sensors, transistors [65]	Tunable properties, high carrier mobility [66]	Immature technology, integration challenges
Calcium Fluoride (CaF ₂)	0.15–9 [67]	~ 1.43 [68]	High thermal stability, low conductivity	Not CMOS compatible	Lenses, prisms, WGs, thermal imaging	Low-loss material, wide transparency range	Brittle, challenging to fabricate for integrated photonics
Other Crystalline Materials	Varies	Varies	Varies (e.g., Sapphire, ZnSe)	Varies	Lenses, prisms, high-power laser applications	Low optical loss, wide transparency range	Brittle, costly, non-CMOS compatible

fragility, and challenges in harsh environments, which may limit their practical utility.

Mashanovich et al. presented propagation loss measurements at the longest wavelength yet reported for an SOI platform [78]. The propagation losses of the WG were as low as 0.6 – 0.7 dB/cm , which can be achieved at a wavelength of $3.39 \mu\text{m}$. Additionally, the propagation loss data for Si on porous Si (SiPSi) WGs at the same wavelength was reported, offering further insight into

alternative platforms for long-wavelength photonics. The cross sectional view of a SiPSi WG produced through large-area proton beam irradiation is shown in Figure 2a [79]. Penades et al. introduced a set of fundamental photonic components utilizing suspended Si WGs, supported by a lateral cladding made of subwavelength grating metamaterials [80]. Their work covers the design, fabrication, and performance evaluation of key devices, including WG bends, multimode interference devices, and Mach-Zehnder interferometers (MZIs), all operating in the 3.7 – $3.8 \mu\text{m}$,

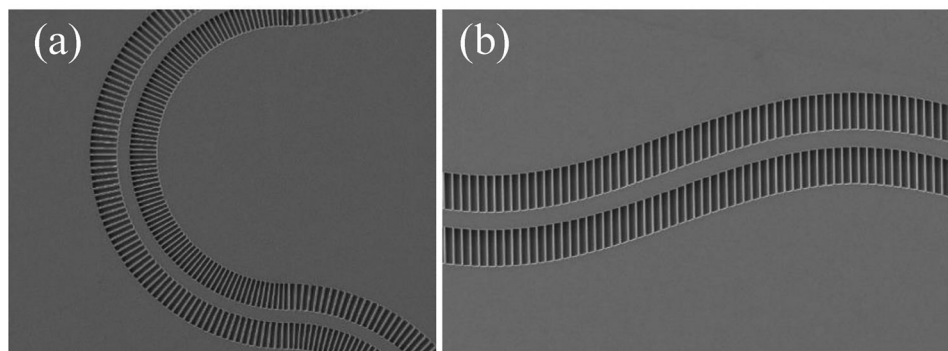


FIGURE 2 | SEM image of (a) 90-degree bend [80], (b) S-bend [80].

marking the first demonstration of such devices on this platform (Figure 2b,c). A WG propagation loss of 0.82 dB/cm was achieved, representing one of the lowest losses reported for Si WGs in this wavelength range [80].

As an alternative, the silicon-on-sapphire (SOS) platform has gained attention due to its wide transparency of up to 5.5 μm and a high RI contrast of approximately 1.7 between Si and sapphire [81, 82]. This large index contrast facilitates strong optical confinement and compact device designs [83]. Despite these advantages, SOS and other emerging platforms each have their limitations, and a variety of alternative material systems have been proposed for MIR applications [84, 85]. These include platforms with different strengths and weaknesses, underscoring the ongoing efforts to optimize MIR Si photonics for diverse applications [81, 86].

2.2 | SiN and SiC

SiN is emerging as a promising platform for integrated photonics, offering features that have contributed to the success of the more mature SOI technology [87, 88]. Key among these is its RI contrast between the core and oxide cladding ($\Delta n \sim 0.5$), which facilitates tight light confinement and supports high-density integration. Additionally, SiN is fully compatible with standard CMOS processes, leveraging advancements in the microelectronics industry to reduce costs at scale. SiN presents distinct advantages over Si in certain applications. While it offers a lower RI contrast, this reduces scattering losses and mitigates two-photon absorption at NIR wavelengths, leading to low-loss devices and high-Q resonators. The lower contrast also increases tolerance to fabrication imperfections [89]. Moreover, SiN's nonlinear properties, such as parametric amplification and broadband supercontinuum generation, combined with its transparency in the NIR and visible spectrum, open new avenues for CMOS-compatible photonics [90]. This technology is particularly well-suited for biosensing applications in the visible and NIR range, where its low losses and reduced sensitivity to thermal variations are critical advantages [55, 91]. As such, SiN is poised to become a leading platform for future integrated photonics solutions [92, 93].

PICs utilizing lithographically patterned WGs provide a scalable and robust platform for manipulating photonic bits, enabling the development of complex and stable photonic technologies [94]. These advances have supported a wide range of applications,

from high-speed optical transceivers to neuromorphic computing and terabit-scale communication. However, next-generation applications demand even lower power consumption and higher operating frequencies [95]. To meet these requirements, SiC, a third-generation semiconductor, presents a promising alternative to traditional Si-based platforms [89]. SiC offers a unique combination of properties, including a wide bandgap, high optical nonlinearities, a high RI, controllable artificial spin defects, and compatibility with CMOS fabrication processes [96]. These exceptional characteristics make SiC an ideal candidate for advancing PIC technology. However, achieving efficient light confinement through total internal reflection presents a challenge, particularly when SiC is grown as thin films on substrates with a higher RI, such as Si. A promising solution proposed by Garrisi et al. involved fabricating suspended subwavelength WGs using a single lithography step [97]. This approach not only addresses the light confinement issue but also maintains the design versatility essential for building a scalable and integrated photonic platform. Recent developments in SiC-based nanophotonics have led to the creation of innovative devices, such as WGs, microcavities, nonlinear frequency converters, and optically active spin defect systems [98].

The incorporation of low-loss SiN platforms with low-temperature materials, including Si electronic and photonic devices, III-V compound semiconductors, lithium niobate (LN), organic materials, and glasses, has been limited by the necessity for high-temperature annealing and the intricacy of disparate processing flows for thin and thick WGs. To enable the next generation of 3D SiN integration, innovative methodologies are required to maintain exceptional low-loss performance, nonlinear optical properties, and CMOS compatibility, while addressing these challenges [99–101].

Bose et al. exhibited a significant advancement in SiN-integrated photonics, achieving record-low optical losses using an anneal-free fabrication process with a maximum temperature of 250 $^{\circ}\text{C}$ [102]. This approach utilized a deuterated silane-based fabrication flow for both SiN and oxide, supporting a wide range of nitride thicknesses without the need for stress mitigation or post-process polishing. The results demonstrated exceptionally low losses for both the nitride core and oxide cladding, achieving a loss of 1.77 dB/m and a Q-factor of 14.9 million for 80 nm nitride core WGs. This represented a more than fivefold reduction in losses compared to previously reported sub-300 $^{\circ}\text{C}$ processes. For 800 nm-thick nitride, a loss as low as 8.66 dB/m and a Q factor

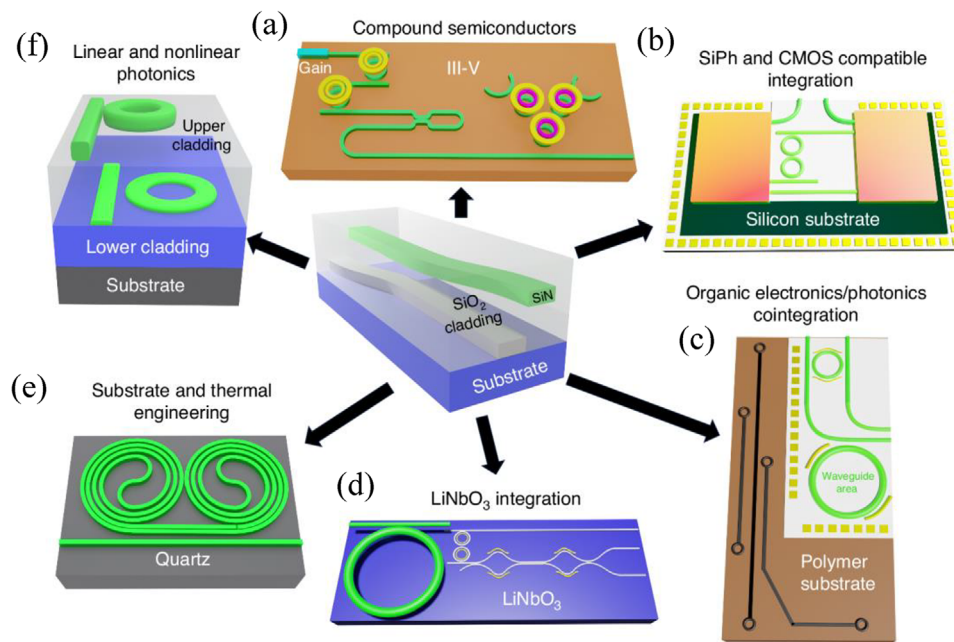


FIGURE 3 | Integration with various technologies for high-performance applications, including (a) compound semiconductors for lasers, (b) pre-processed Si circuits and photonics, (c) organic electronics and photonics, (d) thin film LN, (e) thermal and substrate engineering (e.g., quartz), and (f) combining thick (>650 nm) and thin nitride core devices for different uses [102].

of 4.03 million was reported, the highest reported Q-factor for low-temperature processed resonators of comparable size. The median values for loss and Q-factor across these devices are 13.9 dB/m and 2.59 million, respectively.

Furthermore, several key functionalities enabled by this platform were demonstrated. These included laser stabilization with greater than four orders of magnitude reduction in frequency noise using a thin nitride reference cavity, and in thick nitride micro-resonators, optical parametric oscillation (OPO), two-octave supercontinuum generation, four-wave mixing, and parametric gain were observed. Notably, the lowest optical parametric oscillation threshold per unit resonator length to date was demonstrated. These results represented a transformative step toward the development of a unified, ultra-low-loss SiN platform. This platform supports both linear and nonlinear PICs and is fully compatible with low-temperature materials and processes, facilitating seamless heterogeneous and monolithic integration for a broad range of applications [102].

Bose et al. provided examples of potential heterogeneous and monolithic integration made possible by their anneal-free process. These include depositing ultra-low-loss WGs on III-V semiconductors [103] (Figure 3a) for high-performance lasers and PICs (Figure 3b), integrating with pre-processed electronic circuits and Si photonics, and using organic materials for cointegration with SiN PICs and biophotonics [104] (Figure 3c). The process also worked with thin film LN [105] (Figure 3d) and materials like quartz for stabilizing resonators [106] (Figure 3e). Additionally, it supported the creation of advanced SiN PICs, integrating thin-WG components like Brillouin lasers [107] and thick-WG components for optical frequency combs [108] (Figure 3f).

2.3 | Ge and Ge-on-Si

Ge is an attractive material for MIR-integrated photonics due to its wider transparency range, extending from 2 to 14 μm [46]. Ge has a higher RI than Si, which enhances the light confinement in WGs and allows for efficient guiding of MIR wavelengths [109]. Ge-on-Si platforms leverage Si's mature fabrication infrastructure while extending the operational range to longer wavelengths than those supported by Si alone [110, 111]. In particular, Ge WGs have progressed for use in sensing applications that rely on the detection of chemical and biological signatures in the MIR [112]. Ge-based photonic devices have also been shown to enable nonlinear optical processes, making them suitable for frequency conversion applications. Additionally, the integration of Ge with Si photonics allows for the development of hybrid platforms that can simultaneously operate in both NIR and MIR regimes.

Change et al. successfully demonstrated a Ge strip WG on a Si substrate, optimized for single-mode transmission of TM polarized light at a wavelength of 5.8 μm , generated by QCLs [113]. Using the Fabry-Perot resonance method, a minimum propagation loss of 2.5 dB/cm was measured. Additionally, a bending loss of just 0.12 dB for a 90° bend with a 115 μm radius was reported. Furthermore, Nedeljkovic et al. presented transmission measurements of Ge-on-Si WGs in the 7.5–8.5 μm wavelength range, achieving a minimum propagation loss of 2.5 dB/cm at 7.575 μm [60]. Notably, an unexpected and significant increase in propagation loss at higher wavelengths was observed. In addition, the first Ge-on-Si multimode interferometers designed for operation in this spectral range were demonstrated, along with grating couplers optimized for use with long-wavelength IR cameras. Furthermore, the “cut-back” method for loss measurement was employed, enabling simultaneous transmission measurements through multiple WGs of varying lengths. High-quality optical end-facet preparation

was achieved using ductile regime dicing, offering a fast and reproducible process for WG end-face optimization [60].

In another instance, Anantha et al. demonstrated a Ge-based platform with a high core-cladding index contrast, utilizing Ge-on-SiN WGs for MIR applications [114]. To validate the design, simulations were conducted, confirming the structure's feasibility. The WG is fabricated by bonding a SiN-deposited Ge-on-SiN donor wafer onto a Si substrate, followed by a layer transfer process, enabling scalability across different wafer sizes. Misfit dislocations at the Ge-Si interface, initially present after bonding, are eliminated through chemical-mechanical polishing, resulting in a high-quality Ge layer. At a MIR wavelength of 3.8 μm , the Ge-on-SiN WG exhibits a propagation loss of 3.35 ± 0.5 dB/cm and a bending loss of 0.14 ± 0.01 dB for a 5 μm radius in the transverse-electric mode.

Xiao et al. experimentally demonstrated a Ge nanocavity in the MIR range with a Q-factor of approximately 18,000 which is the highest reported for nanocavities within MIR Ge-based platforms to date [115]. This breakthrough was achieved through a combination of sophisticated theoretical design, advanced Smart-Cut wafer fabrication techniques, and carefully optimized device manufacturing processes. Figure 4a presents an SEM image of the fabricated device, while Figure 4b provides a close-up view of the central region, specifically the photonic crystal nanobeam cavity. With its high Q-factor and ultra-small mode volume, the nanocavity paves the way for innovative on-chip applications within the MIR spectrum [115].

2.4 | Chalcogenide Glasses

Chalcogenide glasses, which are compounds of chalcogen elements such as sulfur, selenium, or tellurium with metals like arsenic or Ge, offer a highly promising platform for MIR photonics [116]. These glasses have excellent optical transparency across the entire MIR spectrum, from 2 to 12 μm , and can be engineered to exhibit low optical losses in this range due to molecular vibrations of atmospheric gases such as H_2O , CO_2 , and O_3 . However, the optical characteristics of these glasses vary depending on their composition. Glasses based on sulfur, for instance, show partial transparency in the visible spectrum (Figure 4c) [61], while those incorporating selenium or tellurium are typically opaque. While it is relatively straightforward to extend the IR transmission to longer wavelengths by incorporating heavier chalcogen elements, which alter phonon vibrations, expanding transparency into shorter wavelengths is far more challenging due to its dependence on the electronic bandgap. To address this, understanding the microstructure of the glasses is essential, as it directly influences the gap between the valence and conduction bands.

Their high RI and strong nonlinear optical properties make chalcogenide glasses suitable for various purposes, including optical switching, nonlinear optics, and MIR waveguiding [117]. Chalcogenide glasses are also highly compatible with standard microfabrication techniques, enabling the fabrication of integrated PICs. Their soft nature compared to Si and other crystalline materials makes them easier to process and shape into complex structures like rib and strip WGs, resonators, and optical

fibers. One drawback of chalcogenide materials, however, is their relatively low thermal and mechanical durability, which can limit their use in high-power functions [118, 119].

In 2007, Hu et al. introduced the first microfluidic device monolithically integrated with planar $\text{Ge}_{23}\text{Sb}_7\text{S}_{70}$ chalcogenide channel WGs, enabling the detection of N-methylaniline through the absorption fingerprint of its N-H bond at approximately 1496 nm [120]. Building on this, the development of a microfluidic sensor based on Ge-Sb-Se glass was subsequently advanced to detect phenylethylamine molecules, which exhibit absorption at a wavelength of 1550 nm [121]. To enhance sensitivity, optical resonators integrated with planar microfluidic systems have been constructed, although these operated primarily in the NIR [122]. More recently, high-Q chalcogenide glass-on-silicon resonators and photonic crystal cavities, operating at 5.2 μm , have been put forward as a solution [123, 124]. However, despite these advances, none of the chalcogenide (ChG) WG sensors have fully leveraged the MIR transparency of chalcogenides or exploited the strong fundamental molecular vibrational absorption bands in the MIR. Ideally, for molecular differentiation, devices should operate across the functional group and fingerprint regions, spanning 2.5–11 μm ($4000\text{--}900\text{ cm}^{-1}$). However, the limited availability of WGs in this spectral range has restricted demonstrations of chemical and biochemical sensing to date [125].

A novel MIR WG gas sensor, based on ChG material deposited on a MgF_2 substrate, was fabricated using the lift-off technique [126]. The MgF_2 layer served as the lower cladding, enhancing the external confinement factor to improve light-gas interaction. CO_2 gas detection was carried out using wavelength modulation spectroscopy (WMS) at a wavelength of 4319 nm (2315.2 cm^{-1}). The sensor, with a 1 cm long sensing WG, achieved a detection limit of approximately 0.3%, representing an over 8-fold improvement compared to sensors utilizing direct absorption spectroscopy (DAS). The integration of WMS with this WG-based sensor introduces a novel approach for enhancing the sensitivity and performance of on-chip gas detection systems [126].

Ma et al. presented the characteristics of low-loss chalcogenide WGs designed for MIR sensing purposes [116]. The WGs feature a $\text{Ge}_{11.5}\text{As}_{24}\text{Se}_{64.5}$ rib core coated with a 10 nm fluoropolymer layer, supported by a $\text{Ge}_{11.5}\text{As}_{24}\text{Se}_{64.5}$ bottom cladding. Fabrication was achieved through thermal evaporation, photolithography, and ICP plasma etching techniques. Across most of the functional group region, spanning 1500 to 4000 cm^{-1} , the WGs exhibit propagation losses below 1 dB/cm, with a minimum loss of 0.3 dB/cm at 2000 cm^{-1} . The WG's potential for MIR spectroscopy was demonstrated by successfully measuring the absorption spectrum of soluble Prussian blue in dimethyl sulfoxide.

2.5 | III-V Semiconductors

III-V semiconductors, such as GaAs [127] and indium phosphide (InP) [128], are ideal for MIR photonics because of their direct bandgap, high electron mobility, and ability to be engineered with precise bandgap tunability [129]. These materials exhibit strong optical absorption and emission in the MIR range (3–8 μm), making them well-suited for purposes like gas sensing, IR imaging, and telecommunications. Their ability to form heterostructures

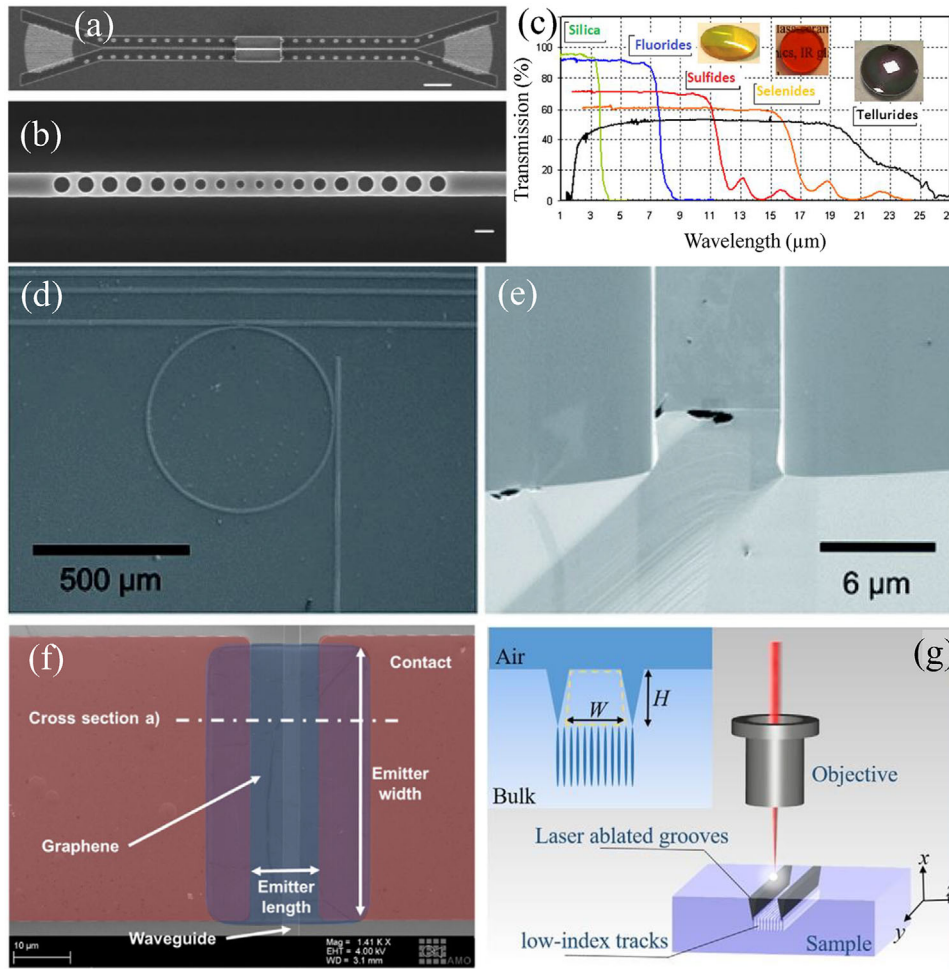


FIGURE 4 | SEM images of the MIR Ge device, showing the high-Q nanocavity (a) Top view [115], (b) Close-up of the high-Q nanocavity, with a scale bar of 500 nm [115], (c) Transparency window of S, Se, Te-based chalcogenide glasses. Insets show examples of sulfide As_2S_3 (yellow), selenium As_2Se_3 (red), and tellurium $\text{Te}_{20}\text{As}_{30}\text{Se}_{50}$ (black) glasses [61], (d) GaAs μ -RR with alignment, in-coupling, and de-coupling WGs [140], (e) Angled top and front view of a $6 \times 6 \mu\text{m}$ tilted GaAs WG on a $6 \mu\text{m}$ $\text{Al}_{0.2}\text{Ga}_{0.8}\text{As}$ buffer layer [140], (f) SEM image of the thermal emitter [151], (g) Schematic of the fabrication process using a femtosecond (fs) laser. The inset shows a diagram of the ridge structure [160].

and quantum wells also enhances device performance by allowing for greater control over light-matter interactions. Moreover, III-V semiconductors can be integrated with other photonic technologies, offering versatility for designing efficient MIR lasers and detectors [130, 131].

The advanced technological development of InGaAs/AlInAs/InP QCLs (QCLs) has spurred significant interest in creating MIR PICs based on InP [132]. Both intrinsic $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ and InP semiconductor compounds offer a wide transparency window in the MIR range. Additionally, the capability to achieve extremely low background doping levels (below 10^{15} cm^{-3}) in these materials, either by minimizing impurities or utilizing doping compensation techniques, enables absorption losses in bulk structures to remain well under 1 dB/cm [133–135]. Zhang et al. demonstrated MIR ring resonators (RRs) fabricated on an $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}/\text{InP}$ material platform, operating at a wavelength of approximately $5.2 \mu\text{m}$ and achieving quality factors exceeding 120,000 [128]. Additionally, a comprehensive broadband analysis of optical losses was conducted in passive $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}/\text{InP}$ ridge WGs across the 5–11 μm wavelength range. The measured

WG losses range from below 0.5 dB/cm at shorter wavelengths to 4–5 dB/cm at the longer wavelengths within this spectrum. These findings confirm $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}/\text{InP}$ as a low-loss platform for MIR PICs, which are fully compatible with InGaAs/AlInAs/InP quantum cascade lasers and detectors [128].

Karnik et al. presented the development of low-loss passive WGs and highly efficient arrayed WG gratings, suitable for purposes such as beam-combining IR laser arrays [136]. The WG design incorporated an $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ core surrounded by InP cladding layers, selected for their compatibility with potential monolithic integration alongside QCLs. Various PICs were manufactured using standard semiconductor processes, and experimental results from these devices revealed WGs with an estimated propagation loss of approximately 1.2 dB/cm and μ -RRs with an intrinsic Q-factor of 174,000. The arrayed WG gratings, operating within the 5.15–5.34 μm wavelength range, exhibited minimal insertion loss (IL) of ~ 0.9 dB and low non-uniformity (~ 0.6 dB). This work highlighted significant progress toward the monolithic integration of compact IR light sources with advanced functionalities, offering promising

potential for applications in chemical sensing and high-power systems.

In the realm of nonlinear crystals with significant second-order susceptibilities, ideal for user-friendly frequency conversion via Quasi-Phase Matching (QPM), the introduction of Periodically Poled Lithium Niobate (PPLN) and similar oxides has achieved sustained commercial success. More recently, Orientation-Patterned semiconductors have been developed to meet the demands of longer-wavelength applications and are frequently compared to PPLN. For free-space beams with diameters smaller than approximately 1 mm, the primary practical distinction among the crystals is their respective transparency windows. For many applications, bulk Orientation-Patterned Gallium Arsenide (OP-GaAs) is often regarded as the “PPLN of the MIR,” covering wavelengths from below 2 μm to approximately 15 μm [137, 138]. Additionally, advanced thick orientation-patterned growth techniques have been applied to other semiconductor materials.

Gallium antimonide (GaSb), with its high second-order susceptibility of 150 pm/V and thermal conductivity of 32 W/m.K, presents itself as a strong candidate for QPM in nonlinear integrated optics. Roux et al. explored the low-loss OP-GaSb/AlGaAsSb WGs specifically engineered for parametric conversion for the first time, targeting pump wavelengths around 2 μm and extending into the MIR range, which is highly valued for its spectroscopic applications [139]. μ -RRs featuring a decoupling WG were fabricated from GaAs/Al_{0.2}Ga_{0.8}As, designed for MIR wavelengths, and characterized through thermal tuning as shown in Figure 4d. In Figure 4e, the front view of a cleaved GaAs WG facet (6 $\times$ 6 μm), deposited on a 6 μm -thick Al_{0.2}Ga_{0.8}As buffer layer, is displayed. The image reveals a clean, well-defined end facet with minimal scattering losses, indicating high-quality fabrication. These devices demonstrated a Q-factor of 1900, a thermal FWHM of 8 $^{\circ}\text{C}$, and a thermal FSR of 18 $^{\circ}\text{C}$ [140]. The relatively low Q-factor suggests an efficient coupling between the input and decoupling WGs. These μ -RRs are promising candidates for advanced MIR chemical and biological sensing applications through resonant-cavity enhancement. Additionally, they offer high spectral resolution, making them well-suited for evanescent field sensing and effective wavelength de-multiplexing in WG-based systems [140].

2.6 | 2D Materials

2D materials have attracted significant interest due to their exceptional properties, including a tunable bandgap, high carrier mobility, and adjustable absorption and response across various spectral bands [65]. These unique characteristics position 2D materials as highly promising candidates for a wide range of advanced IR (IR) technologies, such as night vision, remote sensing, surveillance, target acquisition, and optical communication [141]. Gaining a deeper understanding of the potential and performance of 2D materials in IR applications is therefore critical for the development of next-generation high-performance devices in these fields [142].

Since the discovery of graphene in 2004, 2D materials have attracted substantial research interest due to their unique and

highly tunable electrical and optical properties [143]. Recent studies have further revealed the potential of various 2D materials in MIR photonics and optoelectronics [144]. Graphene, characterized by its linear Dirac-like bandstructure [145], supports ultra-broadband optical absorption, spanning from the visible to the far-IR range. Beyond graphene, other 2D materials such as black phosphorus and TMDs have also been widely explored for MIR optoelectronic applications. Compared to traditional bulk MIR materials like HgCdTe, 2D materials exhibit remarkable advantages, including ultrafast carrier dynamics, broadband spectral responses, compatibility with 2D heterostructures, and CMOS-based integration [146, 147]. Numerous 2D material-based photonic and optoelectronic devices have recently been demonstrated, including PDs, modulators, polaritonic devices, and nonlinear optical systems, generating significant interest across various scientific fields such as materials science, photonics, applied physics, and electrical engineering [142, 148–150].

Negm et al. presented graphene-based MIR emitters integrated with photonic WGs, capable of coupling directly into the fundamental mode of Si WGs optimized for the “fingerprint region,” which is crucial for gas sensing applications (Figure 4f) [151]. These WG-integrated graphene emitters exhibited high broadband emission intensity. Successful coupling into the WG mode was confirmed by the emission detected at the output grating couplers. Thermal simulations suggested emitter temperatures reaching up to 1000 $^{\circ}\text{C}$, where blackbody radiation effectively spans the MIR range. The coupling efficiency, defined as the ratio of light emitted into the WG to the total emission, was estimated at up to 68%, significantly surpassing the performance of other WG-integrated emitters reported in the literature [151].

The thermo-optic bistability in a graphene-coated Si₃N₄ RR was investigated by Gao et al., demonstrating significant performance enhancements [152]. By precisely controlling the monolayer graphene coverage on the resonator, a two-fold increase in the resonance shift rate driven by thermo-optic effects was observed, as well as an 18-fold enhancement in the effective thermal nonlinear RI compared to devices lacking graphene. Additionally, the thermo-optic hysteresis loop within this hybrid structure was characterized, with experimental results aligning closely with theoretical predictions. These findings open new avenues for developing graphene-on-Si₃N₄-based high-speed PDs, modulators, and other integrated devices for nonlinear optical applications on chip [152].

In recent years, 2D materials such as transition metal dichalcogenides (TMDs) have gained attention for their potential in MIR photonics [24, 148]. TMDs, such as molybdenum disulfide (MoS₂) [153] and tungsten diselenide (WSe₂) [154], exhibit unique electronic and optical properties that make them promising for use in MIR devices. These materials can be easily exfoliated into atomically thin layers, which can be integrated into photonic structures to create novel devices such as modulators, detectors, and tunable filters [155]. TMDs are transparent over a wide range of MIR wavelengths and can be engineered to exhibit strong light-matter interactions, leading to enhanced optical nonlinearities. This makes them attractive for nonlinear optical devices such as frequency converters and parametric amplifiers. Although still in the research phase, TMDs hold great promise for the future of MIR-integrated photonics, especially in applications

that require extreme miniaturization and high-performance light manipulation [66].

Emerging 2D van der Waals semiconductors offer a powerful platform for heterostructure formation, enabling precise control over bandgap engineering for advanced atomically thin photonic applications. When these heterostructures exhibit a type-II band alignment, interlayer excitons (ILEs) can form due to rapid charge transfer across the layers. Edelstein et al. introduced a MIR PD and modulator integrated with a WG, exploiting ILE absorption [156]. The device was built from a heterostructure consisting of a monolayer of tungsten disulfide (WS_2) and several layers of hafnium disulfide (HfS_2), seamlessly integrated with a Si WG on a sapphire substrate. Broadband MIR photodetection across the 3.8–5.5 μm range was observed, achieving responsivity on the scale of tens of $\mu\text{A/W}$, with minimal impact on WG transmission. Moreover, all-optical modulation was demonstrated in the MIR range by manipulating the ILE population via interband transitions in the individual heterostructure layers [156].

The development of low-cost, easily integrable PDs for Si photonics remains a significant challenge, particularly for wavelengths exceeding 1.8 μm . Multilayer platinum diselenide (PtSe_2), a semi-metallic 2D material, offers a promising solution due to its ability to be synthesized at temperatures below 450 $^\circ\text{C}$. Parhizkar et al. successfully integrated PtSe_2 -based PDs through direct conformal growth on Si WGs [157]. These PDs demonstrated excellent performance at a 1550 nm wavelength, with a peak responsivity of 11 mA/W and rapid response times of less than 8.4 μs . Fourier-transform IR spectroscopy over a broad wavelength range (1.25 to 28 μm) further highlights the suitability of PtSe_2 for IR detection applications. The direct-growth approach significantly outperformed conventional PtSe_2 PDs produced via 2D layer transfer methods. With its combination of IR sensitivity, chemical stability, selective low-temperature growth, and potential for high carrier mobility, PtSe_2 emerged as a highly attractive 2D material for optoelectronic devices and PICs [157].

2.7 | CaF_2 and Other Crystalline Materials

Crystalline materials like CaF_2 , magnesium fluoride (MgF_2), and LN are well-known for their excellent transparency in the MIR range [50, 158]. In the context of integrated photonics, crystalline materials like CaF_2 are being explored for the fabrication of MIR WGs and resonators [158]. These materials exhibit low optical losses and high damage thresholds, making them suitable for high-power and high-performance applications in the MIR spectrum [159]. One limitation of crystalline materials is the difficulty of integrating them with established photonic platforms such as Si. However, advances in microfabrication techniques, such as ion implantation and wafer bonding, have opened new possibilities for integrating these materials into MIR PICs. For instance, CaF_2 -based resonators have been demonstrated for use in MIR frequency comb generation and sensing applications.

The fabrication process of ridge WGs in CaF_2 crystals designed for MIR applications was investigated by Zhang et al. [158]. Initially, planar WGs were created through O^{5+} ion irradiation, establishing the foundational structure. Subsequently, the ridge WG geometry was achieved by employing a precise diamond

blade dicing technique. Propagation loss was evaluated using an end-face coupling method, after which annealing was performed to enhance WG performance. This thermal treatment successfully reduced the propagation loss to 0.5 dB/cm at an operational wavelength of 4 μm . To assess potential lattice damage induced by O^{5+} ion irradiation, Raman spectroscopy was employed on both the WG and the substrate, providing insights into the structural integrity of the material.

Fs laser micromachining presents a highly effective and versatile approach for fabricating advanced photonic components in transparent optical materials, enabling a wide range of applications [161–164]. Li et al. introduced a novel design and fabrication method for ridge WGs within LiNbO_3 crystals, optimized for MIR operation, using an all-fs-laser processing technique [160]. Figure 4g illustrates the fs laser micromachining process. A focused fs laser pulse induced localized modifications within the focal volume, resulting in the formation of low-index cladding tracks and laser-ablated grooves. These outcomes were influenced by various laser parameters, including pulse energy, repetition rate, pulse duration, scanning speed, and beam polarization. The inset in Figure 4g provides a schematic representation of the ridge WG structure, where W represents the bottom width of the ridges, and the ablation depth is approximately $H \sim 25 \mu\text{m}$. These ridge WGs demonstrated efficient light guidance at a wavelength of 4 μm . Furthermore, Y-branch WG structures, serving as 1×2 beam splitters, were successfully fabricated, achieving near-equal splitting ratios ($\sim 1:1$) at the same wavelength [160].

3 | Applications

MIR integrated photonics is an emerging field with significant potential across a diverse range of applications (Figure 5), driven by the unique properties of light in the MIR spectrum. This spectral range corresponds to the vibrational absorption bands of many molecules, making it particularly well-suited for precise chemical identification and analysis. Consequently, MIR photonics is finding transformative applications in areas such as chemical and environmental sensing, where real-time monitoring of gases and pollutants is crucial, and in biomedical fields for non-invasive diagnostics. In sectors like agriculture and the food industry, MIR techniques are used for quality control and contamination detection. The field also supports laser-based manufacturing processes, offering new capabilities in precision material processing, and contributes to energy applications, such as in the development of more efficient thermophotovoltaic systems. This diverse array of applications underscores the growing significance of MIR photonics in modern technological landscapes, as discussed in this section.

3.1 | Chemical and Environmental Sensing

Gas sensing is one of the most prominent applications of MIR photonics, leveraging the fact that many gases have unique absorption fingerprints in the MIR spectrum [165]. Molecules like carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), water vapor (H_2O), sulfur dioxide (SO_2), carbon monoxide (CO), ammonia (NH_3), ozone (O_3) and various volatile organic compounds (VOCs) exhibit strong vibrational and rotational transitions in

Applications

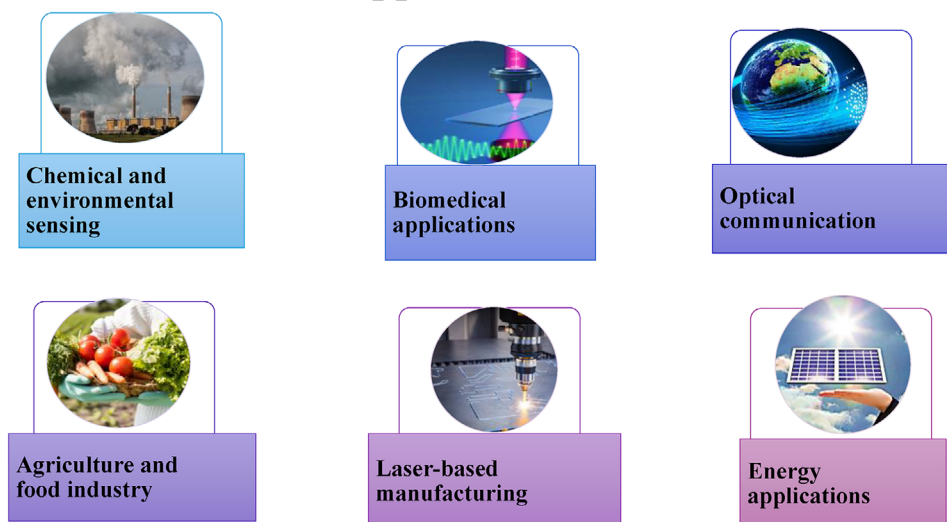


FIGURE 5 | Potential applications of MIR integrated photonics.

this wavelength range, which allows highly sensitive and selective detection [166]. Integrated photonics in the MIR spectrum is revolutionizing this field by offering miniaturized, robust, and scalable solutions for gas detection.

In traditional gas sensing, bulky systems such as Fourier-transform infrared (FTIR) spectrometers are often used to detect gases based on their absorption spectra. However, with integrated photonics, key components such as light sources (QCLs, ICLs), WGs, and detectors can be fabricated on a single chip using materials tailored for MIR wavelengths, like chalcogenide glass, Si, or Ge. These compact, chip-scale systems allow for real-time and on-site gas sensing with high precision and without the need for large laboratory equipment. For example, a MIR photonic sensor can emit a laser at a specific wavelength corresponding to the absorption line of a target gas and measure the decrease in intensity as the light passes through the gas sample. By integrating tunable laser sources and spectrometers, these systems can also detect multiple gases simultaneously, making them suitable for applications in industrial process monitoring, environmental compliance, and safety systems. One of the key benefits of using integrated photonics in the MIR range for gas sensing is the enhancement of sensitivity and resolution [167–169]. The strong light-matter interactions in the MIR allow for the detection of gases at very low concentrations, even in the parts-per-billion (ppb) range. Several novel gas sensing devices based on evanescent field absorption have been recently proposed [168, 170–175].

Environmental monitoring using MIR-integrated photonics is becoming an increasingly important tool in efforts to address global climate change, pollution control, and ecosystem management [176, 177]. The MIR spectral range provides access to key absorption bands of a wide variety of atmospheric pollutants and greenhouse gases, including CH_4 , N_2O , SO_2 , and O_3 . These gases contribute significantly to the greenhouse effect, acid rain, and other environmental issues, making their precise monitoring crucial for regulatory compliance, climate research, and indus-

trial monitoring. For instance, CH_4 , a potent greenhouse gas, has strong absorption lines around $3.3\ \mu\text{m}$ [169], making it an ideal target for MIR sensing technologies. Similarly, SO_2 and CO_2 can be monitored using MIR light in the $4\text{--}8\ \mu\text{m}$ range [178]. In integrated photonic sensors, MIR light is guided through specialized WGs that interact with the surrounding environment, where the presence of a gas is detected by measuring the specific absorption at the corresponding wavelength [179].

Ranacher et al. introduced a novel IR evanescent-field absorption gas sensor, engineered with a Si strip WG (1 cm long) specifically tailored for CO_2 detection as shown in Figure 6a [180]. Utilizing finite element simulations, the WG design was optimized to enhance sensitivity and performance. To optimize functionality, a taper and launch pad were integrated at both ends of the WG. Each launch pad features a grating coupler, which facilitates the coupling of MIR radiation from an external MIR source into the WG and directs the out-coupled light to a detector. Figure 6b illustrates a schematic of the experimental setup, including a gas cell housing the test chip. Experimental results demonstrated the sensor's capability to perform accurate quantitative measurements of CO_2 concentrations, with detection thresholds as low as 5000 ppm. This sensitivity aligns with the occupational exposure limits established in various regulatory standards [180].

A chip-scale MIR sensor was developed specifically for detecting hydrocarbon gases. The sensor utilized amorphous silicon (a-Si) optical ridge WGs, fabricated using CMOS technology [8]. To assess the sensing capabilities of the a-Si WG, C_2H_2 and CH_4 were selected for evaluation. The probe light wavelength was scanned across the 3.00 to $3.40\ \mu\text{m}$ range, encompassing the characteristic C–H absorption bands for both gases. Figure 6c illustrates the captured WG mode images as various gases were introduced into the PDMS chamber. When N_2 was injected, the mode intensity remained strong throughout the scan due to the IR inactivity of the $\text{N}\equiv\text{N}$ bond. However, when C_2H_2 was introduced, a noticeable reduction in light intensity occurred between 3.00

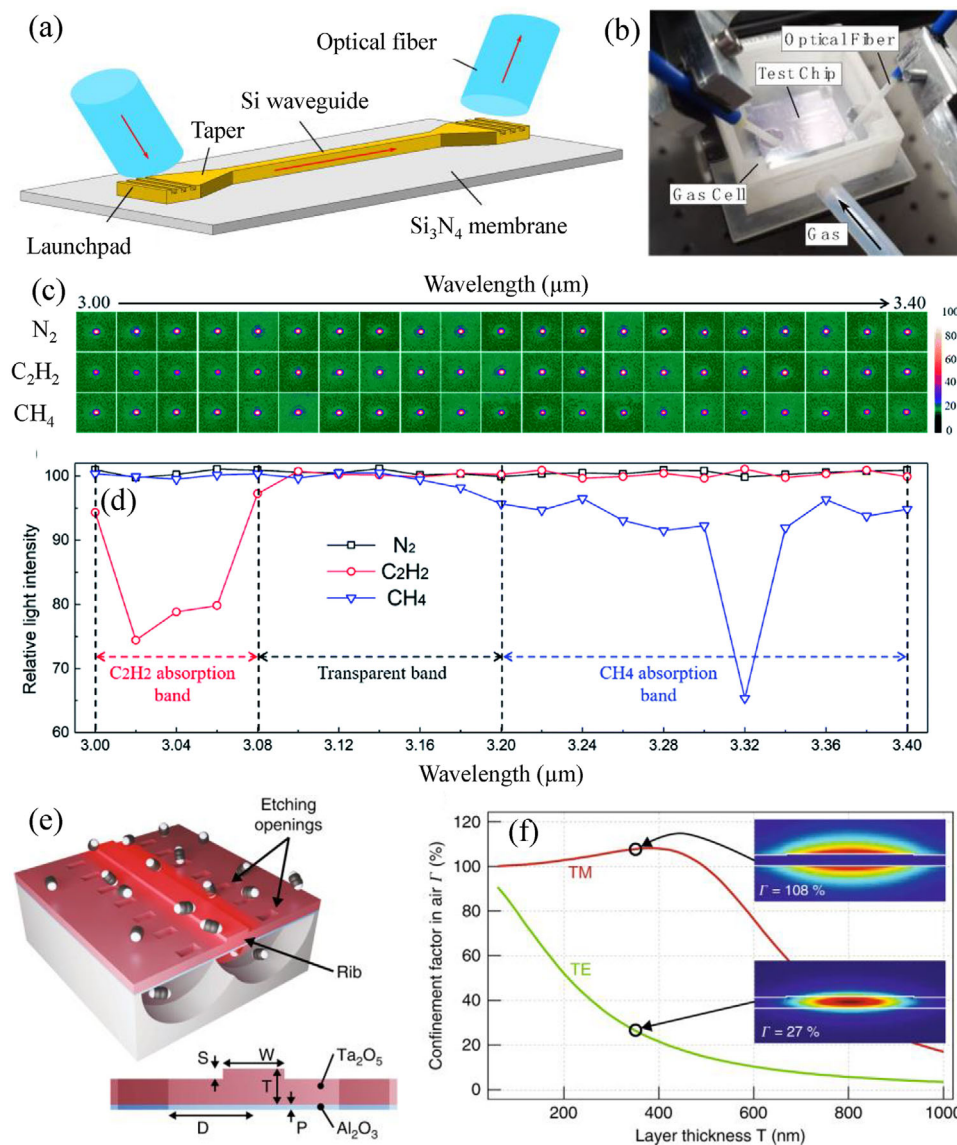


FIGURE 6 | (a) Schematic diagram of the measurement setup [180], (b) The gas cell containing the test chip and connected optical fibers [180], (c) Images of the WG modes captured as N_2 , C_2H_2 , and CH_4 gases were introduced into the PDMS chamber, with wavelengths scanned from 3.0 to 3.4 μm [8], (d) Mode intensity results show significant attenuation between 3.00–3.08 μm for C_2H_2 and 3.29–3.33 μm for CH_4 [8], (e) Schematic of the free-standing shallow rib WG interacting with C_2H_2 molecules. The WG is made from a Ta_2O_5 core layer, passivated underneath by Al_2O_3 , with under etching through openings [39], (f) Simulated air confinement factor vs. Ta_2O_5 thickness (T). Insets show the $|\mathbf{E}|^2$ field distributions for the fundamental TM and TE modes, along with the air confinement factor for $T = 350$ nm [39].

and 3.08 μm , after which the mode brightened. In contrast, when CH_4 was injected, the mode intensity was consistent across most of the spectrum, except for a significant dip around 3.32 μm . To provide a more detailed analysis, the WG mode intensity was recorded at specific wavelengths and is shown in Figure 6d. For C_2H_2 , substantial attenuation was observed between 3.00 and 3.08 μm , attributed to its asymmetric C–H stretching. In the case of CH_4 , the mode intensity dropped between 3.18 and 3.40 μm , corresponding to the asymmetric C–H vibration, ν_3 . The absorption between 3.32 and 3.40 μm was identified as the strong R branch of CH_4 , while the absorption from 3.18 to 3.30 μm was linked to the milder P branch. These results confirmed the ability of the MIR WG sensor to effectively distinguish between C_2H_2 and CH_4 based on their unique C–H absorption spectra [8].

Vlk et al. presented a MIR-integrated suspended membrane WG sensor that achieves a 107% evanescent field confinement in air, surpassing free-space optics in optical interaction per unit length, as shown in Figure 6e [39]. Additionally, the sensor's design minimizes facet reflections, leading to a flat spectral baseline and exceptionally low absorbance noise, making it competitive with free-space spectroscopy. When tested at 2.566 μm , the sensor demonstrated a detection limit of 7 ppm for acetylene using a WG just 2 cm in length. The field distribution perpendicular to the membrane surface was primarily determined by the membrane's thickness. For TM polarization, the evanescent field fraction increases significantly as the membrane thickness decreases. At a thickness of $T = 500$ nm, even minimal WG dispersion was enough to raise the external confinement factor (Γ) beyond 100% as shown in Figure 6f. In the case of Ta_2O_5 , which has a

moderate RI, Γ peaks at 108% when the membrane reaches a thickness of around 400 nm. This indicates that a robust per-length interaction can be achieved with a relatively thick yet mechanically stable membrane [39].

3.2 | Biomedical Applications

Biomedical diagnostics is one of the most promising applications of MIR-integrated photonics due to the ability of MIR light to interact strongly with molecular vibrations in biological tissues and fluids [181, 182]. In this spectrum, biomolecules like proteins, lipids, carbohydrates, and nucleic acids exhibit distinct absorption patterns, enabling label-free and highly specific detection of these molecules [40]. Integrated photonic systems capitalize on these properties to develop compact, non-invasive diagnostic tools with high sensitivity and molecular specificity [182].

One key application of MIR photonics is biofluid analysis, where biological samples such as blood, saliva, or urine are analyzed to detect biomarkers for diseases [183, 184]. MIR light probes these samples, revealing the presence and concentration of specific biomolecules based on their characteristic absorption peaks. For example, glucose exhibits strong absorption in the MIR range, allowing for the development of non-invasive glucose monitoring devices for diabetes patients [185]. Unlike traditional monitors requiring blood samples, MIR sensors can perform the measurement using sweat or even directly through the skin, enhancing patient comfort and compliance. This technology could also be used to detect cancer or inflammatory biomarkers by identifying molecular changes in body fluids, enabling early diagnosis of conditions such as cancer or cardiovascular diseases [186].

Koyama et al. proposed a MIR spectroscopic system utilizing a high-speed wavelength-swept, pulsed QCL for biomedical applications, particularly non-invasive blood glucose measurement [41]. The system employed an attenuated total reflection (ATR) measurement setup, which incorporates a QCL equipped with a micro-electromechanical system (MEMS)-scanning grating, hollow-core optical fibers, and an indium arsenide antimonide (InAsSb) detector. Its feasibility for healthcare diagnostics was evaluated. The system generated a continuous spectrum by integrating multiple comb-shaped spectra, each of which is temporally shifted. This approach allowed real-time acquisition of absorption spectra without the need for complex computational processing. It was observed that the signal-to-noise ratio (SNR) of the spectra improved with an increase in the number of spectra integrated during the calculation. Consequently, the absorption spectrum of a 0.1% aqueous glucose solution was successfully recorded. Additionally, the system demonstrated the capability to measure the absorption spectra of human lip tissue. Using partial least squares (PLS) regression analysis, a model equation was developed that enabled the estimation of blood glucose levels based on the measured absorption spectra. The suggested QCL-based spectroscopic system, featuring a MEMS-scanning grating, offered notable advantages in terms of compactness and cost-effectiveness compared to conventional FTIR spectrometers and typical QCL-based systems that rely on large, mechanically tuned gratings. These benefits made it a promising solution for portable and affordable healthcare diagnostics [41].

Another cutting-edge application of MIR-integrated photonics is tissue diagnostics, where the absorption characteristics of biological tissues are used for non-invasive imaging and molecular fingerprinting. IR spectroscopy based on MIR light can identify biochemical changes in tissues, such as those associated with cancer. For instance, cancerous tissues undergo molecular transformations, such as altered lipid-to-protein ratios, which are reflected in their absorption spectra. Integrated MIR photonic devices can be miniaturized for clinical use, providing rapid and detailed tissue analysis as an alternative to traditional histopathology. These systems can also be used in real-time during surgery, ensuring complete removal of cancerous tissue and improving outcomes [187].

In therapeutic applications, MIR photonics is crucial for laser-based surgery and ablation. Lasers in this spectrum, such as erbium-doped or QCLs, have wavelengths that are strongly absorbed by water and other biological molecules, making them ideal for precise tissue removal with minimal thermal damage. Lasers around 3 μm , for example, are highly absorbed by water, making them effective for cutting or vaporizing soft tissues like skin or mucous membranes without excessive heat buildup, reducing scarring and accelerating healing [188]. MIR photonics is also valuable in ophthalmic surgery, where precise tissue control is critical. For instance, corneal reshaping procedures like LASIK utilize MIR lasers to ablate thin layers of the cornea, correcting vision with fine control over the depth of tissue removal [189]. This technology can be applied to minimally invasive surgeries in other areas, such as soft tissue ablation for cancer treatment, where tumors can be removed with minimal damage to surrounding healthy tissue [190].

In addition to surgery, MIR-integrated photonics is used in therapies such as photothermal therapy. Here, MIR light selectively heats and destroys cancerous cells. Tumors have different absorption properties than healthy tissues, enabling targeted heating to destroy the tumor without harming surrounding areas. This non-invasive technique can be delivered via optical fibers in endoscopic procedures, making it especially useful for treating hard-to-reach tumors. MIR photonics also contributes significantly to medical imaging, particularly in non-invasive and real-time diagnostics [191]. Thermal imaging, which captures IR radiation emitted by tissues, can be enhanced using MIR photonics to develop compact, high-resolution cameras and sensors for clinical use. Thermal imaging is particularly useful for distinguishing inflammatory conditions, infections, or abnormal tissue growth, as these often have different thermal profiles compared to healthy tissues. Moreover, MIR photonics is integrated into hyperspectral imaging systems, combining imaging and spectroscopy to analyze tissue molecularly. This technology shows promise in detecting skin cancers by identifying subtle biochemical changes in tissues, allowing for earlier and more accurate diagnoses [192].

A breakthrough in mid-infrared photothermal (MIP) imaging has shown potential for high-resolution 2D and 3D single-cell imaging by leveraging the spatial precision of visible light. Despite this, the technique has been constrained by a maximum measurement rate of just a few frames per second, restricting its broader applicability. Ishigane et al. presented a major improvement in wide-field MIP quantitative phase microscopy, achieving a signal-

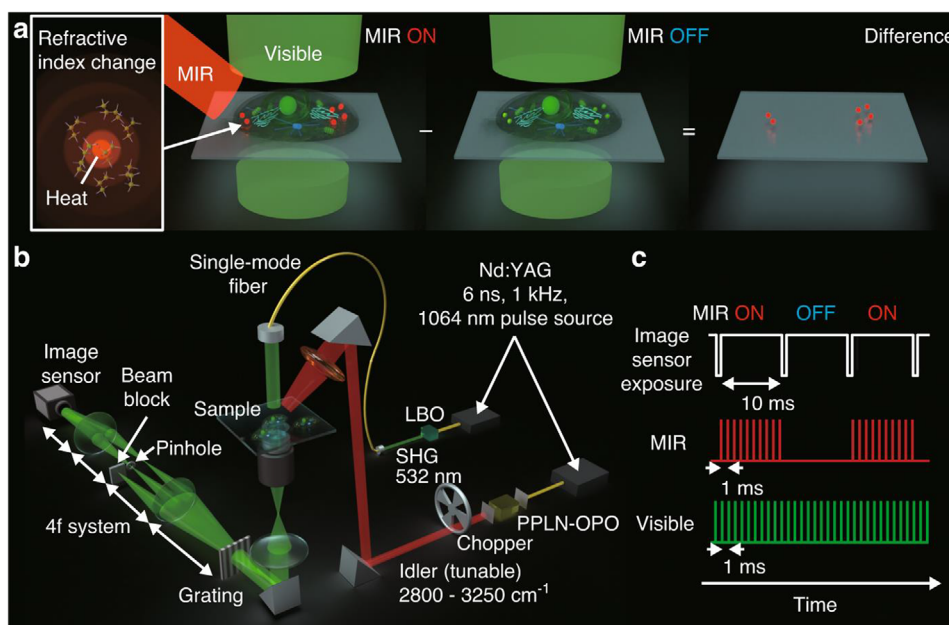


FIGURE 7 | (a) Overview of MIP-QPI principles [42], (b) High-SNR MIP-QPI optical system [42], (c) Timing diagram for MIR and visible pulses [42].

to-noise ratio two orders of magnitude higher than previous methods [42]. This advancement enabled live-cell imaging at speeds surpassing video rates. As shown in Figure 7a, a MIP image is produced by subtracting the phase images taken in the MIR-ON and MIR-OFF states. Among the various MIP recognition methods, quantitative phase imaging (QPI) is the most effective for measuring intracellular molecular distributions. Unlike phase-contrast microscopy, which introduces artefacts like halos, or dark-field and interferometric scattering (iSCAT) microscopes that lose spatial-frequency data, QPI avoids these issues and provides accurate MIP images. Additionally, the phase image in the MIR-OFF state offers detailed morphology and dry-mass information, enabling correlation between target molecule distribution and cellular structures. Figure 7b illustrates the schematic of the MIP-QPI system developed by Ishigane et al. [42]. MIR light was intensity-modulated at 50 Hz, synchronized with the image sensor, alternating between MIR-ON and -OFF frames (Figure 7c). The MIR beam was loosely focused onto the sample with an $\sim 80 \mu\text{m}$ spot size, while the visible beam irradiated with a peak fluence of $\sim 30 \text{ pJ}/\mu\text{m}^2$. Common-path off-axis digital holography was employed as QPI, where a diffraction grating separated the transmitted light into object and reference waves. Interference fringes were captured by a high-full-well-capacity sensor, and the phase image was numerically reconstructed. The system achieved a spatial resolution of 440 nm and is resistant to speckle noise due to the low-coherent visible light used in the differential measurement. This new high-speed, high-resolution MIR microscope shows great promise as a powerful tool for live-cell analysis and other life science applications [42].

Finally, MIR photonic systems were developed for real-time monitoring of drug delivery and pharmacokinetics. By tracking the interaction between MIR light and specific drugs, it became possible to monitor drug metabolism and distribution in the body in real-time. This capability was especially valuable in personalized medicine, where treatments can be tailored based on a

patient's response. MIR-integrated photonics, with its compact and sensitive design, offers an ideal platform for wearable or portable devices that continuously monitor therapeutic efficacy [193].

3.3 | Optical Communication

Recent developments in optical communication using MIR-integrated photonic devices have made significant strides, driven by the demand for higher bandwidth, lower power consumption, and new applications like environmental sensing and secure communication [1, 89, 194–196]. Advances in material science have been key, with new platforms such as SiGe, chalcogenide glasses, and group IV semiconductors allowing for efficient light manipulation at MIR wavelengths [3, 197]. These materials provide the necessary transparency and thermal stability required for MIR communication, outperforming conventional Si photonics used in NIR systems [198].

A critical breakthrough has been the development of integrated lasers, modulators, and detectors specifically designed for MIR wavelengths, enabling compact, high-performance communication systems on a chip. Innovations in QCLs [199] and ICLs [200, 201] have improved the efficiency and output power for MIR light sources, while advances in WG design, such as suspended and hollow-core structures, have reduced propagation losses and enhanced signal transmission distances [3, 156, 202]. Furthermore, the integration of MIR devices with CMOS technology has progressed, enabling mass production and lowering costs. Researchers have also explored new encoding schemes and multiplexing techniques to enhance data transmission rates over MIR bands [16]. These developments are pushing MIR photonic devices toward applications in free-space optical communication, fiber-based networks, and secure, long-range communication systems, particularly in challenging environments where MIR

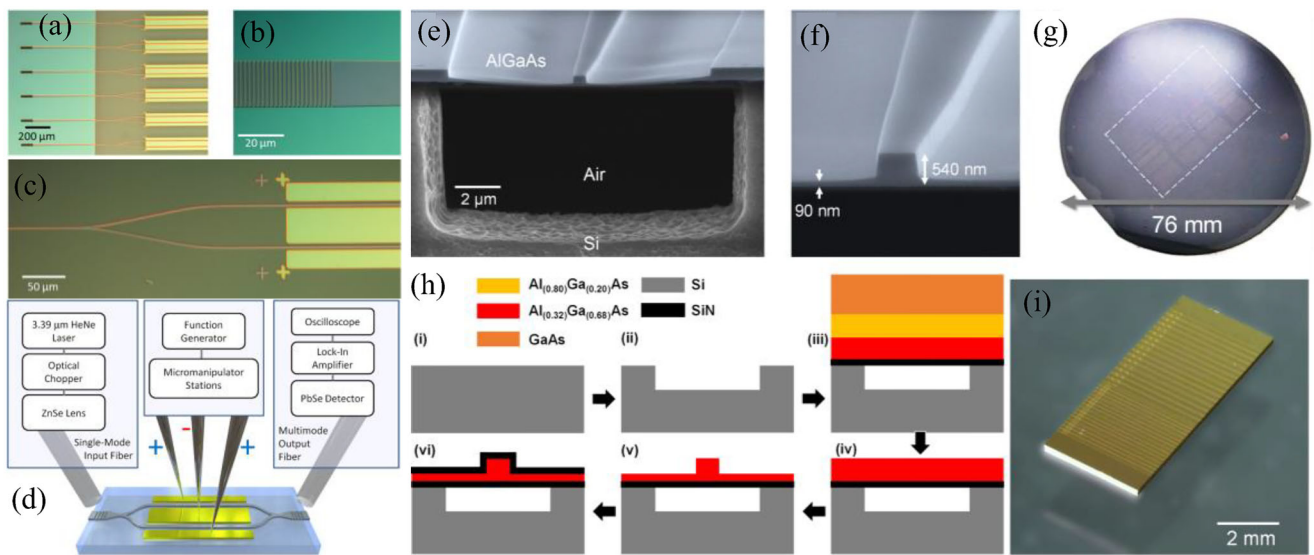


FIGURE 8 | Optical images of a SiLN modulator chip: (a) wide view of the chip showing half-etched (left) and fully-etched (right) areas [205], (b) fiber-to-WG grating coupler [205], and (c) Y-junction and modulator electrodes [205], (d) measurement setup [205]. Suspended AlGaAs on a Si platform: (e) and (f) show SEM images of a typical WG facet formed through dry etching [203]; (g) a photograph of the processed wafer before die release, where an AlGaAs film covers the entire surface but devices were fabricated only in the central dashed region to reduce e-beam lithography time [203]; (h) fabrication process flow summarized in six key steps [203]; (i) focus-stacked image of a μ -RR die following its release from the wafer [203].

wavelengths can outperform traditional telecom bands in terms of atmospheric absorption and interference [203].

The utilization of heterogeneous integration techniques, such as direct bonding, has facilitated the resolution of pivotal challenges inherent to integrated photonics [204]. In the expanding field of MIR photonics, advancement has been constrained by the restricted availability of transparent, functional substrates compatible with this wavelength range. A significant advancement in achieving compact, high-performance optical modulation and frequency conversion is the seamless integration of Si photonics with materials that exhibit high second-order nonlinear susceptibility. By transferring large areas of thin, monocrystalline Si onto bulk LN substrates, Chiles et al. created the first Si-based platform capable of exploiting the Pockels effect, or linear electro-optic effect, in the MIR spectrum [205]. This platform enabled the development of integrated Mach-Zehnder interferometer modulators (see Figure 8a–c), which achieved an extinction ratio of approximately 8 dB, a half-wave voltage-length product of 26 V-cm, and an on-chip IL of 3.3 dB at a wavelength of 3.39 μ m (see Figure 8d). Furthermore, the ultrathin optical waveguides fabricated on this platform exhibited a low transverse electric mode propagation loss of 2.5 dB/cm. Looking ahead, this silicon-on-lithium-niobate platform offers promising potential for the development of advanced applications, such as wideband difference frequency generation, which could pave the way for the next generation of integrated mid-infrared light sources [205].

The microscale integration of mid- and long-wave-infrared photonic components presents significant potential for the development of robust, field-deployable chemical sensors and highly efficient IR frequency converters. However, the performance and capabilities of such technologies are intrinsically tied to the choice of material platform, which dictates critical factors such as the strength and type of optical nonlinearities, transparency

range, modal confinement, and physical durability. Later on, Chiles et al. introduced a novel platform based on suspended AlGaAs WGs integrated into Si, exhibiting superior performance across these key metrics [203]. SEM images of a typical WG facet formed through dry etching are shown in Figure 8e,f [203]. A photograph of the processed wafer before die release, where an AlGaAs film covers the entire surface, but devices were fabricated only in the central dashed region to reduce e-beam lithography time (Figure 8g) [203].

The fabrication process is schematically depicted in Figure 8h. Multiple fabrication runs were performed throughout the project, incorporating variations on the primary process flow. The process consists of the following key stages: (i) cleaning the Si handle wafer, (ii) etching trenches to define the locations of the WGs, (iii) bonding an epitaxial III-V wafer to the Si handle, with a SiN passivation layer applied before bonding, (iv) chemically removing the III-V wafer substrate (GaAs) and the etch-stop layer ($\text{Al}_{0.8}\text{Ga}_{0.2}\text{As}$), (v) performing electron-beam lithography and plasma etching to define ridge WG features over the etched trenches, and (vi) completing the process with oxide stripping and SiN surface passivation. Figure 8i shows the focus-stacked image of a μ -RR die following its release from the wafer [161].

Low propagation losses over an extensive wavelength range, spanning nearly two octaves (1.26–4.6 μ m) were demonstrated, with a minimum loss of 0.45 dB/cm at a wavelength of 2.4 μ m. Leveraging the high nonlinearity of this platform, second-harmonic generation was achieved with a 1560 nm pump and demonstrated octave-spanning supercontinuum generation extending to 2.3 μ m using pump pulse energies as low as 3.4 pJ. When pumped in the MIR, the supercontinuum spans from 2.3 to 6.5 μ m. Additionally, the versatility of the platform was showcased through the demonstration of passive MIR devices, including low-loss bends with a radius of 10 μ m, compact power splitters with an efficiency of 96

$\pm 1\%$, and edge couplers with a loss of 3.0 ± 0.1 dB. This platform holds significant promise for the realization of multifunctional integrated photonic systems operating in the MIR regime [203].

3.4 | Agriculture and Food Industry

The agricultural sector has played a pivotal role in shaping the foundations of human civilization, providing essential resources such as food, beverages, clothing, and raw materials for daily life [206]. Despite its significance, traditional farming methods are highly labor-intensive and increasingly insufficient to keep pace with the growing demands of a rapidly expanding population [207]. This imbalance underscores the urgent need for advanced, efficient solutions in crop cultivation and harvesting. In response, cutting-edge technologies, particularly in optics and photonics, have emerged as promising innovations to revolutionize modern agriculture and address these challenges [25, 208].

Food adulteration and authenticity have become growing concerns for consumers and stakeholders across the food production and industry sectors. To address these issues, numerous physicochemical and instrumental analytical methods are currently available for detecting fraud. However, most of these techniques employ a targeted approach, focusing on identifying specific compounds or groups of compounds. These assays tend to be complex and time-consuming, requiring multiple stages of sample preparation and skilled technicians, making them less practical for routine or large-scale applications. Additionally, since many adulterants are unknown, targeted methods often fail to identify them effectively [209].

In recent years, the need for more comprehensive tools has driven research toward the development of untargeted methodologies in food analysis. Techniques such as IR, Raman, nuclear magnetic resonance (NMR) spectroscopy [210], hyperspectral imaging (HSI) [211], and chromatographic methods [212] (e.g., GC-MS and HPLC-MS) offer a molecular fingerprint of the entire food matrix. Among these, MIR spectroscopy has gained prominence as a viable analytical tool, offering an accessible and cost-effective alternative to more complex methods. It is well-suited for use both in production facilities and quality control laboratories. MIR spectroscopy's high-throughput nature provides several advantages, including rapid analysis, minimal or no sample preparation, quick spectral acquisition, and the ability to detect unexpected adulterants or deviations from reference samples [213, 214].

Given the large and intricate spectral datasets produced by MIR spectroscopy, the use of chemometric analysis is essential for extracting relevant information and building predictive models. These models enable exploratory studies, identification of key sample characteristics, and reliable predictions for future sample analyses. MIR spectroscopy operates within the MIR wavelength range primarily due to fundamental molecular vibrations, which can be categorized into four distinct regions: the X–H stretching region (4000 to 2500 cm^{-1}), the triple-bond region (2500 to 2000 cm^{-1}), the double-bond region (2000 to 1500 cm^{-1}), and the fingerprint region (1500 to 600 cm^{-1}). The X–H stretching region corresponds to vibrations in bonds such as O–H, C–H, and N–H. The triple-bond region is associated with the vibrations of $\text{C}\equiv\text{C}$ and $\text{C}\equiv\text{N}$ bonds, while the double-bond region reflects the

vibrations of $\text{C}=\text{C}$, $\text{C}=\text{O}$, and $\text{C}=\text{N}$ bonds. Lastly, the fingerprint region is characterized by complex bending and skeletal vibrations, which provide unique molecular signatures [215]. Table 3 provides a selection of common examples of absorption bands observed in the MIR region.

Recently, there has been growing interest in enhancing the value of underutilized legumes to promote their use in human diets. Acacia seed (AS), a legume with over 40 edible species native to Australia, is one such crop with significant potential. The study conducted by Adiamo et al. aimed to qualitatively assess the chemical composition of 14 commonly consumed AS species from 27 regions across Australia using MIR spectroscopy as a rapid analytical tool. Both raw and roasted AS flour (heated at 180°C for 5, 7, and 9 min) were examined [216]. The ATR-MIR spectra of the analyzed AS samples, along with the corresponding spectral features, are presented in Figure 9 [216]. The MIR spectra display characteristic absorption bands associated with structural and non-structural carbohydrates in the range of 900 – 1240 cm^{-1} , proteins between 1500 – 1700 cm^{-1} , and lipids within the 2800 – 2950 cm^{-1} range. The absorption band near 1744 cm^{-1} is attributed to phosphodiesteres. Additionally, the region from 900 to 1500 cm^{-1} exhibits bands corresponding to aromatic O–H bending, aromatic C–H in-plane bending, and C–O stretching in phenolic compounds. Notable peaks in the spectra include those related to amide-I vibrations (C–O and C–N stretching) around 1641 cm^{-1} , starch at approximately 1045 cm^{-1} , and lipids or fatty acids (aliphatic CH_2 groups) at 2852 cm^{-1} and 2926 cm^{-1} [216]. These findings offered a basis for exploring compositional similarities and differences among AS species before and after roasting, providing insights into their potential for improved food applications.

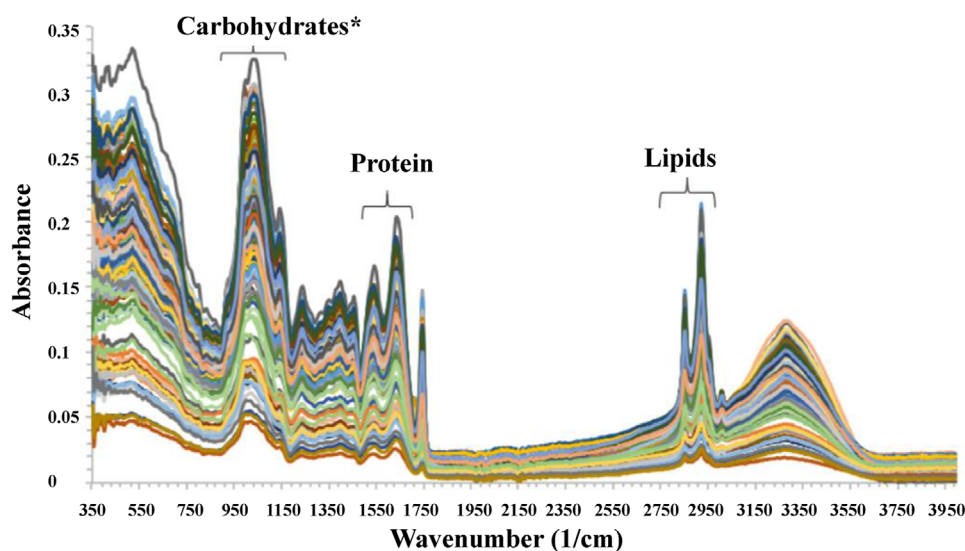
3.5 | Laser-Based Manufacturing

MIR laser-based manufacturing offers an advanced technique that harnesses the distinct properties of MIR light for highly precise material processing [217]. These lasers are well-suited for processing materials like semiconductors, polymers, and glasses commonly used in photonic devices, as they exhibit high absorption in this spectrum. This enables cleaner cuts, reduced thermal damage, and greater accuracy in the fabrication of components like WGs, sensors, and optoelectronic devices [217–219]. MIR lasers are particularly advantageous for producing photonic devices in applications such as sensing, communication, and environmental monitoring, where precision and material integrity are crucial [220, 221].

In the last decade, polymer-based photonic devices have been extensively utilized in various scientific and industrial domains [222, 223]. Polymers offer advantages such as low weight, cost-effectiveness, and robustness. Crucially, their physical properties can be engineered to meet specific functional requirements for diverse applications. Optical WGs, which are fundamental components of photonic devices made from polymers, are typically fabricated through methods like photolithography, hot-embossing, molding, 3D printing, and the mosquito technique [224–227]. Despite this, there have been relatively few demonstrations of WG photo-inscription in polymers. In a recent study, Pätzold et al. described the fabrication of 3D waveguides in poly

TABLE 3 | Notable examples of MIR absorption bands.[215]

Wavelength (μm)	Wavenumber (cm^{-1})	Assignment
Water		
2.778–3.125	3200–3600	O–H stretching
6.061	1650	H–OH stretching
Proteins		
5.917–6.250	1600–1690	Amide I (C=O stretching)
6.349–6.757	1480–1575	Amide II (C–N stretching and N–H bending)
7.692–8.130	1230–1300	Amide III (C–N stretching and N–H bending)
Fats		
3.333–3.571	2800–3000	C–H stretching
5.731–5.797	1725–1745	C=O stretching
10.309	970	C=C–H bending
Carbohydrates		
3.333–3.571	2800–3000	C–H stretching
7.143–12.500	800–1400	Skeletal stretching and bending

**FIGURE 9** | MIR spectra of raw and roasted acacia seed species from various regions of Australia. * Represents structural and non-structural carbohydrates [216].

(methyl methacrylate) (PMMA) using stress-induced structures inscribed with a fs laser. The laser inscription method is advantageous due to its flexibility, enabling the formation of complex geometries in a single-step process [228]. The recent study conducted by Frayssinous et al. has shown that the laser ablation of polymers can be significantly enhanced by tuning a MIR fiber laser to wavelengths that resonate with the fundamental stretching frequency of C–H bonds [229].

Subsequently, Berube et al. proposed an innovative approach based on resonant absorption for the fabrication of on-surface WGs in polymers [230]. By adjusting the MIR lasers to the absorption peaks of PMMA and polycarbonate, the laser beams were focused on the surfaces of bulk samples, thereby inducing

localised modifications in the RI. The laser beam was then scanned rapidly across the surface to process the WGs. The study provided a comprehensive analysis of the photo-induced WGs and evaluated their optical properties, revealing an IL of 3.1 dB for a 30 mm long WG in PMMA, indicating propagation losses of 0.8 dB/cm. Figure 10a,b illustrates the longitudinal (bottom) and cross sectional (top) views of modifications created using various laser fluences. The laser fluence at the centre of the irradiated region was determined by integrating the Gaussian irradiance profile over the laser dwell time. Subsequently, the light intensity distributions transmitted through the WGs were evaluated. The observation of distinct guided modes at the output facet was contingent upon the position of the incident beam relative to the cross sectional plane of the traces. Figure 10c,d depicts the

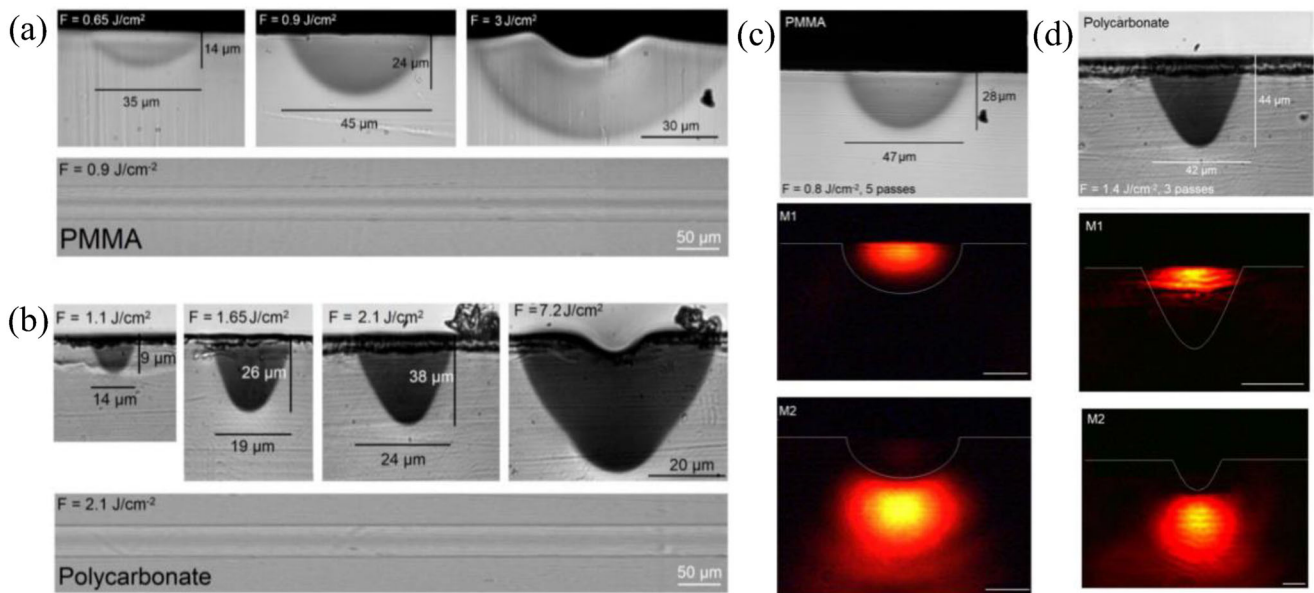


FIGURE 10 | (a) Brightfield images of the cross sectional and longitudinal views of photo-induced modifications created with a single laser pass in (a) PMMA and (b) polycarbonate [230]. The modifications were inscribed using a continuous-wave laser with a wavelength of $3.425 \mu\text{m}$. Near-field intensity profiles of 633 nm light transmitted through WGs inscribed in (c) PMMA and (d) polycarbonate [230].

near-field intensity profiles recorded at the WG output. Two well-defined guiding regions were identified in each of the two polymer materials, which corresponded directly to their RI distributions. The primary mode, designated as M1, exhibited an elliptical shape and propagated from the surface of the polymer to a depth of several microns, and this mode was associated with the presence of an RI gradient in proximity to the surface of the photoinduced modification regions. This is shown in Figure 10.

In addition, a secondary guiding mode, labelled M2, was observed below the photo-induced structure, within the annular region of compressed material. The M2 mode presents a circular geometry with diameters often exceeding $20 \mu\text{m}$. Limited coupling between modes M1 and M2 was detected, with weak power transfer occurring only in larger WGs fabricated using high laser fluence. The irregular and noisy intensity profiles of M1 modes in polycarbonate were attributed to damage along the sample's surface edges. Importantly, these on-surface WGs can function as effective refractometric sensors. As a proof of concept, a small droplet of index-matching oils was deposited onto the WG surfaces, observing attenuations of 23.8 dB and 35.1 dB for liquids that matched the refractive indices of polycarbonate and PMMA, respectively. This demonstrated a substantial increase in sensitivity compared to previously reported optical WG or fiber-based RI sensors [230].

3.6 | Energy Applications

Leveraging an underexplored region of the EM spectrum holds great potential for advancing high-efficiency solar cells, which are critical for ensuring a sustainable energy future [24, 231, 232]. In the millimeter- and micrometer-wave spectrum, rectennas are employed to capture and convert propagating EM radiation into usable energy [233]. Meanwhile, in the UV, visible, and NIR ranges, photovoltaic solar cells are already well-established as

efficient energy harvesters, representing a mature and widely adopted technology [234]. However, in the MIR region, which lies between these well-utilized spectral bands, effective solutions for energy harvesting remain largely undeveloped. A significant portion of the solar spectrum, particularly wavelengths extending to 4000 nm , remains untapped. Moreover, the challenge of capturing and converting approximately 10^{17} W of MIR thermal radiation emitted by the Earth poses a critical opportunity for the renewable energy sector to address [232].

Conventional solar cells and rectennas experience inherent energy losses, primarily due to heat generation and the inability to utilize wavelengths beyond the transparency threshold ($\lambda > 1100 \text{ nm}$ for Si) in the photoelectric conversion process. To address this limitation, a hybrid photovoltaic/thermoelectric system was introduced by Nishijima et al. as an enhancement to standard Si solar cells, enabling energy capture across the full solar spectrum [235]. This system utilized nano-textured Si, also known as black Si, to minimize surface reflectivity, while a Ge-Sn layer is incorporated beneath the cell to improve absorption of IR wavelengths. As a result, the hybrid system achieved up to a 7% increase in voltage output through its combined photoelectric and thermal-to-electrical energy conversion mechanisms [235].

Choi et al. designed a metasurface (MS) for MIR detectors to enhance quantum efficiency (QE) in the 2 to $6 \mu\text{m}$ range [231]. The degree of enhancement was influenced by the material's intrinsic absorption coefficient (α). For instance, with a large α value of $2 \times 10^4/\text{cm}$, a 0.1 -micron-thick MS detector could achieve a peak QE of 90% , a 2.6 -fold improvement over conventional detectors. The enhancement became even more significant with lower α values; at $2000/\text{cm}$, the improvement reached around 10 times, with a peak QE of 49% . For energy harvesting, nanostructures etched onto Si solar cells to boost absorption efficiency were designed. The utilization of an array of nano-columns resulted in a notable increase in absorption, from 55% to 97% at $0.7 \mu\text{m}$

and from 5% to 37% at 1 μm . The incorporation of an array of nano-cones led to a notable enhancement in the average absorption, reaching 95% within the range of 0.4 to 0.8 μm . The integrated absorption exhibited a considerable improvement, amounting to 74% for nano-columns and 92% for nano-cones. In the context of GaAs solar cells, the MS design demonstrated a 26% augmentation in photocurrent compared to a conventional planar cell with a 100 nm-thick absorber [231].

The integration of photonic systems within the MIR spectrum presents substantial potential for numerous advanced applications. Despite black phosphorus (BP) being recognized as a promising platform for diverse active functionalities in the MIR, the realization of integrated modulators remains a critical challenge. Huang et al. reported the successful hybrid integration of an MIR BP modulator with a Si photonic WG [236]. Utilizing the gating effect, the anisotropic absorption characteristics of armchair BP were modulated, enabling efficient optical modulation across the 3.85–4.1 μm wavelength range. The WG design was optimized to enhance light–BP interactions, achieving a modulation depth of approximately 5 dB under a low bias voltage of -4 V. Furthermore, the device exhibited a compact active area of 225 μm^2 and a switching energy of approximately 2.6 pJ, demonstrating significant efficiency improvements compared to conventional modulators. The versatility of this platform was further validated by the integration of both a BP PD and modulator, underscoring the potential of combining 2D materials with Si photonics to develop high-performance, miniaturized optoelectronic devices for on-chip MIR applications [236].

Currently, the development of MIR optical modulator technology remains in its early stages, with only incremental progress. This ongoing challenge continues to act as a significant bottleneck in the advancement of fully integrated on-chip MIR systems (Figure 11a). The integrated modulator incorporates an ITO (indium tin oxide)- Al_2O_3 (aluminium oxide)-BP (black phosphorus) capacitor mounted on the Si WG (Figure 11b). As light propagates through the Si photonics WG, it interacts with the BP layer, and its intensity was modulated by the variation in BP absorption. This absorption change was controlled by adjusting the gate bias between different levels, effectively modulating the optical signal (Figure 11c). The realization of on-chip MIR systems requires the co-integration of key active components, such as PDs and modulators. Figure 11a illustrates a potential application in chemical sensing, where arrays of PDs analyze the optical signal following its interaction with a target chemical, while modulators manipulate the light at distinct frequencies within each channel to improve signal-to-noise differentiation. To advance this technology, a BP modulator and PD were successfully integrated into a single chip (Figure 11d). Under a bias voltage of 0.1 V, the PD demonstrated a responsivity of approximately 0.3 A/W at a wavelength of 4 μm (Figure 11e) [236].

However, it is important to note that even a small, applied bias can induce a significant dark current in the PD, primarily due to the increased channel width resulting from WG integration. To address this, an asymmetric electrode layout was demonstrated, enabling zero-bias operation, which effectively reduces dark current and minimizes standby power consumption. This configuration yielded a current responsivity of approximately 40 mA/W (Figure 11e) and a voltage responsivity of up to 44 V/W (Figure 11f,

at 4 μm). Furthermore, applying a $\pm$ V gate bias to the modulator allows the switching of the open-circuit voltage in the PD, driven by the photovoltaic effect, between two distinct levels (Figure 11g). This integration demonstrated the fabrication compatibility of various active components within a single process flow and validated the functional diversification of the BP/Si photonics platform for MIR applications [236].

4 | Challenges and Prospects

MIR integrated photonics offers substantial potential across diverse sectors, including chemical sensing, environmental monitoring, and medical diagnostics [25]. However, its full potential is constrained by several significant challenges. A primary limitation is the scarcity of efficient and cost-effective light sources, detectors, and materials optimized for this spectral range. Conventional semiconductors, such as Si, exhibit poor transparency in the MIR, necessitating the exploration of alternative materials such as chalcogenides, Ge, or emerging 2D materials [24, 155]. The fabrication of photonic devices utilizing these materials, however, presents significant complexity and high costs, which impede scalability for widespread commercial use. Another critical issue is thermal management, as MIR devices typically generate considerable heat during operation, necessitating advanced thermal regulation technologies [237–239]. Furthermore, the integration of MIR photonic components with established platforms, such as fiber optics and traditional semiconductor technologies, is challenging due to material incompatibilities and other systemic barriers, complicating the development of cohesive, functional systems [240, 241].

Despite these challenges, the prospects for MIR photonics remain highly promising. Recent advancements in QCLs [34, 242] and ICLs [22, 35, 243] have significantly enhanced the performance of MIR light sources. In telecommunications, QCLs demonstrate a significant advantage for data transmission in extreme environments, such as high-temperature or high-pressure conditions, where traditional NIR technologies falter [244]. QCLs operate in the MIR spectrum, offering superior performance in environments with heavy dust, dense gases, or high moisture levels due to their robustness and reduced scattering effects [245]. Unlike NIR technologies, which often suffer from attenuation and reliability issues under such conditions, QCLs ensure consistent, high-fidelity signal transmission, making them invaluable for industrial, aerospace, and remote sensing applications [246].

Ongoing research into novel materials, such as metamaterials and photonic crystals, also holds potential for breakthroughs in device miniaturization and efficiency [247]. Additionally, the increasing demand for portable, compact sensing solutions in fields like environmental monitoring and medical diagnostics is driving innovation in this domain [176]. Emerging applications in free-space optical communications, which offer enhanced data transmission rates and improved security compared to conventional methods, are another key area leveraging the unique properties of MIR wavelengths. Moreover, the integration of artificial intelligence (AI) and machine learning (ML) for MIR spectroscopy data analysis is poised to unlock new capabilities in complex real-time environments, such as chemical detection and pathogen identification [248]. As advancements in materials

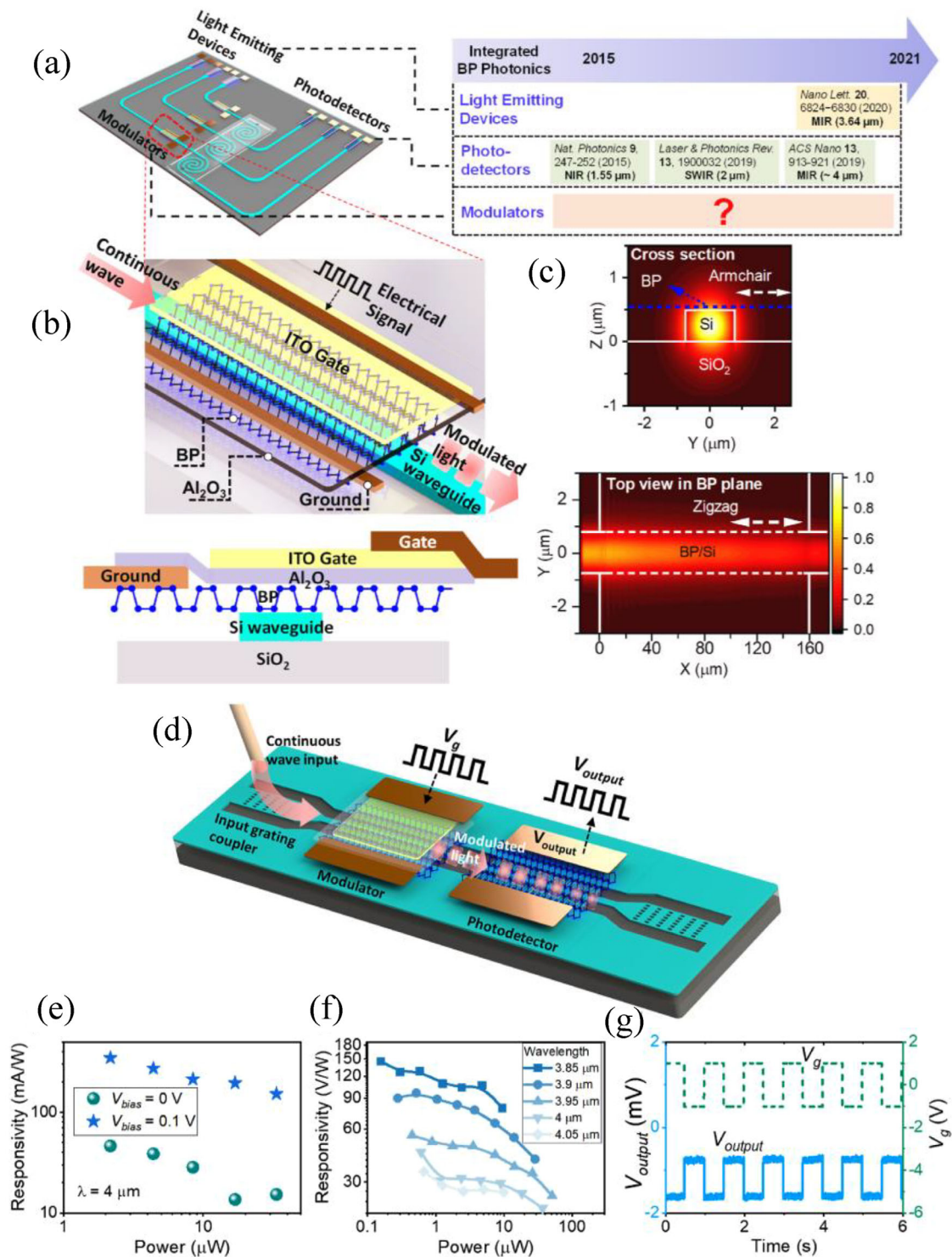


FIGURE 11 | (a) In a MIR system, the key active functions include light emission, modulation, and detection. While this work demonstrates integrated PDs and light-emitting diodes, the previously missing component, a modulator, was addressed, (b) Optical modulation is achieved by applying a vertical electric field to the BP layer, which interacts with the MIR light propagating through the Si WG beneath it. The cross section is perpendicular to the direction of light propagation, (c) As light travels along the Si WG, it is absorbed by the BP in the core region [236], (d) Diagram of the BP modulator and PD integrated on the same chip. A continuous wave is modulated as it travels through the WG below the BP modulator, then directed to the BP PD, (e) Current responsivity and (f) voltage responsivity of the PD is shown, with power dependence due to saturable absorption matching earlier studies. Zero-bias operation helps reduce standby power, (g) The voltage output from the BP PD is shown as the gate bias on the BP modulator switches between +1 and −1 V [236].

science and device engineering progress, MIR photonics is positioned to play an increasingly transformative role across multiple industries, contingent on overcoming existing technological and manufacturing challenges.

To transition MIR integrated photonics from research labs to real-world applications, several key challenges must be addressed. First, the development of standardized, scalable fabrication techniques including wafer-level processing and robust packaging remains crucial [195, 238, 248]. While impressive demonstrations exist, many MIR photonic devices rely on bespoke, lab-scale setups. Achieving CMOS compatibility and reducing fabrication costs will be essential for commercialization. Additionally, the limited availability of efficient, compact, and room-temperature MIR light sources and detectors continues to hamper the deployment of integrated solutions [35, 249, 250]. Overcoming these hurdles will require cross-disciplinary collaboration, including between material scientists, device engineers, and systems integrators.

5 | Final Thoughts

MIR integrated photonics has become a pivotal area of research and development, with significant implications for a wide range of scientific and technological disciplines. This review has systematically examined key material platforms, including chalcogenide glasses, Si, Ge, graphene, and 2D materials, all of which are instrumental in enhancing the performance and functionality of MIR integrated photonic devices. The distinct optical properties of these materials, in combination with advances in fabrication techniques and device miniaturization, have substantially expanded the scope and capabilities of MIR technologies.

The practical applications of MIR photonics in areas such as chemical and biological sensing, environmental monitoring, medical diagnostics, and industrial process control have seen marked improvements in sensitivity and precision. These technologies are particularly valuable in fields such as molecular spectroscopy, free-space communication, and thermal imaging, where they offer enhanced detection capabilities and performance reliability. The diverse applicability of MIR devices underscores their importance in sectors ranging from healthcare to environmental science and telecommunications.

In recent years, numerous cutting-edge and innovative photonic devices have been introduced specifically for MIR applications. These devices leverage advanced technologies and design principles to address the unique challenges of MIR wavelengths, enabling a wide range of applications such as sensing, imaging, communication, and spectroscopy. This paper provides a comprehensive review of these newly proposed photonic devices, highlighting their potential impact, technological advancements, and prospects in the field of MIR photonics.

While significant research progress has been made, the transition of MIR photonic technologies to commercial products remains at an early stage. Some components such as quantum cascade lasers, thermoelectric detectors, and chalcogenide waveguides have reached limited commercial deployment in applications like gas sensing and spectroscopic analysis. However, full system-level

integration, cost-effective packaging, and standardized manufacturing processes are still under development. The current market presence is mostly confined to niche sectors, and widespread adoption will require further advances in reliability, scalability, and regulatory approval, particularly in biomedical and industrial monitoring applications.

Looking forward, the continuous development of material science is expected to further improve the efficiency, integration, and scalability of MIR devices. Such advancements are likely to accelerate the adoption and deployment of MIR photonic technologies in both established and emerging applications. The ongoing investigation and optimization of these material platforms will be essential for fully realizing the potential of MIR integrated photonics, driving innovation and expanding its impact across multiple domains. Furthermore, to translate laboratory advances into widespread real-world adoption, it is critical to address practical challenges beyond research. This includes the development of standardized, scalable fabrication processes compatible with CMOS technology, cost-effective packaging solutions, and robust, room-temperature MIR sources and detectors. Additionally, greater collaboration between academia, industry, and regulatory bodies will be essential to establish clear pathways for commercial integration, particularly in healthcare, environmental sensing, and industrial monitoring. Demonstrating reliable, application-driven systems rather than isolated component performance will be a key step toward unlocking the full potential of MIR integrated photonic technologies.

Acknowledgements

The authors acknowledge financial support from the Chips JU, through the European Union programs Digital Europe (DEP) and Horizon Europe (HE), as well as by the participating states: Spain, Ireland, the Netherlands, Finland, Germany, Portugal, Poland, Austria, Italy, France, and the United Kingdom. In particular, the Grant Agreements of DEP GA101213744 — HE GA101213727 — and the project CJU-010000-2025-2 from the Spanish Government through the Ministerio para la Transformación Digital y de la Función Pública; the Agencia Estatal de Investigación under project Nos. PID2022-139447NB-I00, TED2021-132584B-C21, and PDC2023-145798-I00; the Ministerio de Asuntos Económicos y Transformación Digital under project TSI-069100-2023-0002; and the Generalitat Valenciana under COMCUANTICA 001 & 002.

Conflicts of Interest

The authors declare no conflict of interest.

References

1. T. Hu, B. Dong, X. Luo, et al., “Silicon Photonic Platforms For Mid-Infrared Applications [Invited],” *Photonics Research* 5 (2017): 417–430.
2. S. A. Miller, M. Yu, X. Ji, et al., “Low-loss Silicon Platform for Broadband Mid-Infrared Photonics,” *Optica* 4 (2017): 707–712.
3. H. Lin, Z. Luo, T. Gu, et al., “Mid-Infrared Integrated Photonics on Silicon: A Perspective,” *Nanophotonics* 7 (2018): 393–420.
4. L. Lucia, G. L. Ngo, M. Urbain, et al, in *Integrated Optics: Devices, Materials, and Technologies XXIX* (SPIE, 2025), PC1336900.
5. R. Shankar and M. Lončar, “Silicon Photonic Devices for Mid-Infrared Applications,” *Nanophotonics* 3 (2014): 329–341.

6. C. J. Stirling, M. Nedeljkovic, C. Mitchell, D. J. Rowe, and G. Z. Mashanovich, "Sub-Wavelength Gratings In Silicon Photonic Devices For Mid-Infrared Spectroscopy and Sensing," *Photonics and Nanostructures—Fundamentals and Applications* 58 (2024): 101223.
7. A. Lambrecht and K. Schmitt, in *Mid-Infrared Optoelectronics*, eds. E. Tournié, L. Cerutti (Woodhead Publishing, 2020), 661–715.
8. T. Jin, J. Zhou, and P. T. Lin, "Real-Time and Non-Destructive Hydrocarbon Gas Sensing Using Mid-Infrared Integrated Photonic Circuits," *RSC Advances* 10 (2020): 7452–7459.
9. C. Consani, C. Ranacher, A. Tortschanoff, T. Grille, P. Irsigler, and B. Jakoby, "Mid-Infrared Photonic Gas Sensing Using a Silicon Waveguide and an Integrated Emitter," *Sensors and Actuators B: Chemical* 274 (2018): 60–65.
10. A. Lochbaum, A. Dorodnyy, U. Koch, et al., "Compact Mid-Infrared Gas Sensing Enabled by an All-Metamaterial Design," *Nano Letters* 20 (2020): 4169–4176.
11. M. Le Corvec, F. Charpentier, A. Kachenoura, et al., "Fast and Non-Invasive Medical Diagnostic Using Mid Infrared Sensor," *IRBM* 37 (2016): 116–123.
12. J. Shim, J. Lim, I. Kim, et al., "Room-Temperature Waveguide-Integrated Photodetector Using Bolometric Effect for Mid-Infrared Spectroscopy Applications," *Light: Science & Applications* 14 (2025): 125.
13. M. Nedeljkovic, A. Z. Khokhar, Y. Hu, et al., "Silicon Photonic Devices And Platforms For the Mid-Infrared," *Optical Materials Express* 3 (2013): 1205–1214.
14. G. Z. Mashanovich, L. Reid, G. Georgiev, et al, *Integrated Optics: Devices, Materials, and Technologies XXV* (SPIE, 2022): 1200402.
15. K. Zou, K. Pang, H. Song, et al., "High-Capacity Free-Space Optical Communications Using Wavelength- and Mode-Division-Multiplexing in the Mid-Infrared Region," *Nature Communications* 13 (2022): 7662.
16. A. E. Willner, K. Zou, K. Pang, et al., "Free-Space Mid-IR Communications Using Wavelength and Mode Division Multiplexing," *Optics Communications* 541 (2023): 129518.
17. M. Seminara, T. Gabbriellini, N. Corrias, et al., "Characterization of Noise Regimes in Mid-IR Free-Space Optical Communication Based on Quantum Cascade Lasers," *Optics Express* 30 (2022): 44640–44656.
18. G. Levy, G. F. Koubi, and L. J. Miserendino, "Cutting Efficiency of a Mid-Infrared Laser on Human Enamel," *Journal of Endodontics* 24 (1998): 97.
19. W. Yue, Y. Zhang, L. Shi, et al., "Porcine Skin Ablation Using Mid-Infrared Picosecond Pulse Burst," *Results in Optics* 9 (2022): 100309.
20. M. Razeghi, W. Zhou, S. Slivken, Q.-Y. Lu, D. Wu, and R. McClintock, "Recent Progress of Quantum Cascade Laser Research From 3 to 12 μm at the Center for Quantum Devices [Invited]," *Applied Optics* 56 (2017): H30–H44.
21. K. Pierściński, D. Pierścińska, D. Niewczas, A. Krząstek, A. Broda, and I. Sankowska, "Quantum Cascade Lasers Emitting in the 3.8- μm Range," *Optics & Laser Technology* 192 (2025): 113931.
22. Y. Shen, J. A. Massengale, R. Q. Yang, T. D. Mishima, and M. B. Santos, "Improved Device Performance of Interband Cascade Lasers with Hybrid Cladding Layers Operating in the 3–4 μm Wavelength Region," *Photonics and Nanostructures—Fundamentals and Applications* 57 (2023): 101193.
23. P. G. Schnemann, K. T. Zawilski, L. A. Pomeranz, D. J. Creeden, and P. A. Budni, "Advances in Nonlinear Optical Crystals for Mid-Infrared Coherent Sources," *Journal of the Optical Society of America B* 33 (2016): D36–D43.
24. G. Liang, X. Yu, X. Hu, B. Qiang, C. Wang, and Q. J. Wang, "Mid-Infrared Photonics and Optoelectronics in 2D Materials," *Materials Today* 51 (2021): 294–316.
25. C. J. Mitchell, T. Hu, S. Sun, et al., "Mid-Infrared Silicon Photonics: From Benchtop to Real-World Applications," *APL Photonics* 9 (2024): 080901.
26. D. Chen, S. D. March, A. H. Jones, et al., "Photon-Trapping-Enhanced Avalanche Photodiodes for Mid-Infrared Applications," *Nature Photonics* 17 (2023): 594–600.
27. Y. Nishijima, S. Morimoto, A. Balčytis, et al., "Coupling of Molecular Vibration and Metasurface Modes for Efficient Mid-Infrared Emission," *Journal of Materials Chemistry C* 10 (2022): 451–462.
28. R. Honda, M. Ryu, M. Moritake, et al., "Infrared Polariscopy Imaging of Linear Polymeric Patterns with a Focal Plane Array," *Nanomaterials* 9 (2019): 732.
29. R. Honda, M. Ryu, A. Balčytis, et al., "Paracetamol Micro-Structure Analysis by Optical Mapping," *Applied Surface Science* 473 (2019): 127–132.
30. M. Ryu, R. Honda, A. Balčytis, et al., "Hyperspectral mapping of anisotropy," *Nanoscale Horizons* 4 (2019): 1443–1449.
31. M. Ryu, A. Balčytis, X. Wang, et al., "Orientational Mapping Augmented Sub-Wavelength Hyper-Spectral Imaging of Silk," *Scientific Reports* 7 (2017): 7419.
32. I. Zorin, P. Gatteringer, A. Ebner, and M. Brandstetter, "Advances in Mid-Infrared Spectroscopy Enabled by Supercontinuum Laser Sources," *Optics Express* 30 (2022): 5222–5254.
33. M. A. Butt, M. Juchniewicz, M. Słowikowski, Ł. Kozłowski, and R. Piramidowicz, "Mid-Infrared Photonic Sensors: Exploring Fundamentals, Advanced Materials, and Cutting-Edge Applications," *Sensors* 25 (2025): 1102.
34. A. Schwaighofer, M. Brandstetter, and B. Lendl, "Quantum Cascade Lasers (QCLs) in Biomedical Spectroscopy," *Chemical Society Reviews* 46 (2017): 5903–5924.
35. Y. Deng, Z.-F. Fan, B.-B. Zhao, et al., "Mid-Infrared Hyperchaos of Interband Cascade Lasers," *Light: Science & Applications* 11 (2022): 7.
36. M. Takenaka, Y. Kim, J.-H. Han, J. Kang, and S. Takagi, "Challenges and Opportunities of Near and Mid-Infrared Photonics Based on SiGe and Ge," *ECS Transactions* 75 (2016): 447.
37. E. Sorokin, A. Rudenkov, N. Tolstik, V. Kalashnikov, M. Demesh, and I. T. Sorokina, "Atmospheric Dispersion Management in Mid-IR Mode-Locked Oscillators," *Optics Express* 31 (2023): 18790–18798.
38. P. I. Abramov, A. S. Budarin, E. V. Kuznetsov, and L. A. Skvortsov, "Quantum-Cascade Lasers in Atmospheric Optical Communication Lines: Challenges and Prospects (Review)," *Journal of Applied Spectroscopy* 87 (2020): 579–600.
39. M. Vlk, A. Datta, S. Alberti, et al., "Extraordinary Evanescent Field Confinement Waveguide Sensor for Mid-Infrared Trace Gas Spectroscopy," *Light: Science & Applications* 10 (2021): 26.
40. D. An, F. Sun, Y. Bian, J. Ni, Q. J. Wang, and X. Yu, "Mid-Infrared Absorption Spectroscopy with Enhanced Detection Performance for Biomedical Applications," *Applied Spectroscopy Reviews* 58 (2023): 834–868.
41. T. Koyama, N. Shibata, S. Kino, A. Sugiyama, N. Akikusa, and Y. Matsuura, "A Compact Mid-Infrared Spectroscopy System for Healthcare Applications Based on a Wavelength-Swept, Pulsed Quantum Cascade Laser," *Sensors* 20 (2020): 3438.
42. G. Ishigane, K. Toda, M. Tamamitsu, H. Shimada, V. R. Badarla, and T. Ideguchi, "Label-Free Mid-Infrared Photothermal Live-Cell Imaging Beyond Video Rate," *Light: Science & Applications* 12 (2023): 174.
43. W. Zhou, D. Wu, R. McClintock, S. Slivken, and M. Razeghi, "High Performance Monolithic, Broadly Tunable Mid-Infrared Quantum Cascade Lasers," *Optica* 4 (2017): 1228–1231.
44. Y. Jin, F. Sun, J. Li, et al., "Long Wavelength Mid-Infrared Multi-Gases Spectroscopy Using Tunable Single-Mode Slot Waveguide Quantum Cascade Laser," *Optics Express* 31 (2023): 27543–27552.
45. K. A. Buzaverov, A. S. Baburin, E. V. Sergeev, et al., "Silicon Nitride Integrated Photonics from Visible to Mid-Infrared Spectra," *Laser & Photonics Reviews* 18 (2025): 2400508.

46. D. Marris-Morini, V. Vakarini, J. M. Ramirez, et al., "Germanium-Based Integrated Photonics From Near- to Mid-Infrared Applications," *Nanophotonics* 7 (2018): 1781–1793.
47. X. Lin, M. Wei, K. Lei, et al., "Compact Mid-Infrared Chalcogenide Glass Photonic Devices Based on Robust-Inverse Design," *Laser & Photonics Reviews* 17 (2023): 2200445.
48. J. Haas, R. Stach, C. Kolm, R. Krska, and B. Mizaikoff, "Gallium Arsenide Waveguides as a Platform for Direct Mid-Infrared Vibrational Spectroscopy," *Analytical and Bioanalytical Chemistry* 412 (2020): 3447–3456.
49. C. Gilles, L. J. Orbe, G. Carpintero, G. Maisons, and M. Carras, "Mid-Infrared Wavelength Multiplexer in InGaAs/InP Waveguides Using a Rowland Circle Grating," *Optics Express* 23 (2015): 20288–20296.
50. N.-L. Tran, M. Malerba, A. Talneau, et al., "III-V on CaF₂: A Possible Waveguiding Platform for Mid-IR Photonic Devices," *Optics Express* 27 (2019): 1672–1682.
51. R. Wang, A. Vasiliev, M. Muneeb, et al., "III-V-on-Silicon Photonic Integrated Circuits for Spectroscopic Sensing in the 2–4 μ m Wavelength Range," *Sensors* 17 (2017): 1788.
52. E. Yamasue, M. Susa, H. Fukuyama, and K. Nagata, "Thermal Conductivities of Silicon and Germanium in Solid and Liquid States Measured by Non-Stationary Hot Wire Method with Silica Coated Probe," *Journal of Crystal Growth* 234 (2002): 121–131.
53. M. Shahbaz, M. A. Butt, and R. Piramidowicz, "Breakthrough in Silicon Photonics Technology in Telecommunications, Biosensing, and Gas Sensing," *Micromachines* 14 (2023): 1637.
54. L. Y. Beliaev, E. Shkondin, A. V. Lavrinenko, and O. Takayama, "Optical, Structural and Composition Properties of Silicon Nitride Films Deposited by Reactive Radio-Frequency Sputtering, Low Pressure and Plasma-Enhanced Chemical Vapor Deposition," *Thin Solid Films* 763 (2022): 139568.
55. S. Romero-García, F. Merget, F. Zhong, H. Finkelstein, and J. Witzens, "Silicon Nitride CMOS-Compatible Platform for Integrated Photonics Applications at Visible Wavelengths," *Optics Express* 21 (2013): 14036–14046.
56. M. A. Butt, Ł. Kozłowski, M. Golas, et al., "Numerical and Experimental Demonstration of a Silicon Nitride-Based Ring Resonator Structure for Refractive Index Sensing," *Applied Sciences* 14 (2025): 6082.
57. M. Naftaly, J. F. Molloy, B. Magnusson, Y. M. Andreev, and G. V. Lanskii, "Silicon Carbide—A High-Transparency Nonlinear Material for THz Applications," *Optics Express* 24 (2016): 2590–2595.
58. A. Boretto, Q. Li, and S. Castelletto, "Pioneering the Future with Silicon Carbide Integrated Photonics," *Optics & Laser Technology* 181 (2025): 111910.
59. M. Nedeljkovic, R. Soref, and G. Z. Mashanovich, "Predictions of Free-Carrier Electroabsorption and Electrorefraction in Germanium," *IEEE Photonics Journal* 7 (2015): 2600214.
60. M. Nedeljkovic, J. S. Penades, V. Mittal, et al., "Germanium-on-Silicon Waveguides Operating at Mid-Infrared Wavelengths up to 85 μ m," *Optics Express* 25 (2017): 27431.
61. L. Calvez, "Chalcogenide Glasses and Glass-Ceramics: Transparent Materials in the Infrared for Dual Applications," *Comptes Rendus. Physique* 18 (2017): 314.
62. K. Vyas, D. H. G. Espinosa, D. Hutama, et al., "Group III-V Semiconductors as Promising Nonlinear Integrated Photonic Platforms," *Advanced Physics: X* 7 (2022): 2097020.
63. T. I. Alam, K. Liu, S. U. Hani, S. Ahmed, and Y. H. Tsang, "Advances in Group-10 Transition Metal Dichalcogenide PdSe₂-Based Photodetectors: Outlook and Perspectives," *Sensors* 24 (2024): 6127.
64. X. Huang, C. Zhi, Y. Lin, et al., "Thermal Conductivity of Graphene-Based Polymer Nanocomposites," *Materials Science and Engineering: R: Reports* 142 (2020): 100577.
65. K. Wu, Y. Wang, C. Qiu, and J. Chen, "Thermo-Optic All-Optical Devices Based on Two-Dimensional Materials," *Photonics Research* 6 (2018): C22–C28.
66. R. Cao, S. Fan, P. Yin, et al., "Mid-Infrared Optoelectronic Devices Based on Two-Dimensional Materials Beyond Graphene: Status and Trends," *Nanomaterials* 12 (2022): 2260.
67. https://www.tydexoptics.com/materials/for_transmission_optics/caf2/.
68. I. H. Malitson, "A Redetermination of some Optical Properties of Calcium Fluoride," *Applied Optics* 2 (1963): 1103–1107.
69. M. A. Butt, B. Janaszek, and R. Piramidowicz, "Lighting the Way Forward: The Bright Future of Photonic Integrated Circuits," *Sensors International* 6 (2025): 100326.
70. J. K. S. Poon, A. Govdeli, A. Sharma, et al., "Silicon Photonics for the Visible and Near-Infrared Spectrum," *Advances in Optics and Photonics* 16 (2024): 1–59.
71. M. W. Pruessner, N. F. Tyndall, J. B. Khurgin, W. S. Rabinovich, P. G. Goetz, and T. H. Stievater, "Broadband Near-Infrared Emission in Silicon Waveguides," *Nature Communications* 15 (2024): 4639.
72. R. Soref, "Mid-Infrared Photonics in Silicon and Germanium," *Nature Photonics* 4 (2010): 495–497.
73. A. Govdeli, J. C. Mikkelsen, E. Bebeti, et al., *CLEO 2025 (2025), Paper SS181_2* (Optica Publishing Group, 2025), SS181_2.
74. J. Chiles, S. Khan, J. Ma, and S. Fathpour, "High-Contrast, All-Silicon Waveguiding Platform for Ultra-Broadband Mid-Infrared Photonics," *Applied Physics Letters* 103 (2013): 151106.
75. J. S. Penadés, A. Sánchez-Postigo, M. Nedeljkovic, et al., "Suspended Silicon Waveguides for Long-Wave Infrared Wavelengths," *Optics Letters* 43 (2018): 795–798.
76. J. S. Penadés, C. Alonso-Ramos, A. Z. Khokhar, et al. "Suspended SOI Waveguide with Sub-Wavelength Grating Cladding for Mid-Infrared," *Optics Letters* 39 (2014): 5661–5664.
77. Q. Qiao, H. Sun, X. Liu, et al., "Suspended Silicon Waveguide With Sub-Wavelength Grating Cladding for Optical MEMS in Mid-Infrared," *Micromachines* 12 (2021): 1311.
78. G. Z. Mashanovich, M. M. Milošević, M. Nedeljkovic, et al., "Low Loss Silicon Waveguides for the Mid-Infrared," *Optics Express* 19 (2011): 7112–7119.
79. G. Z. Mashanovich, W. R. Headley, M. M. Milosevic, et al., *7th IEEE International Conference on Group IV Photonics* (IEEE, 2010), 374–376.
80. J. S. Penades, A. Ortega-Monux, M. Nedeljkovic, et al., "Suspended Silicon Mid-Infrared Waveguide Devices with Subwavelength Grating Metamaterial Cladding," *Optics Express* 24 (2016): 22908–22916.
81. T. Baehr-Jones, A. Spott, R. Ilıc, et al., "Silicon-on-Sapphire Integrated Waveguides for the Mid-Infrared," *Optics Express* 18 (2010): 12127–12135.
82. V. Sumanth, S. A. C. Manish, and R. K. Sonkar, *2020 IEEE 8th International Conference on Photonics (ICP)* (IEEE, 2020), 103–104.
83. Y. Zou, S. Chakravarty, and R. T. Chen, "Mid-Infrared Silicon-on-Sapphire Waveguide Coupled Photonic Crystal Microcavities," *Applied Physics Letters* 107 (2015): 081109.
84. F. Li, S. D. Jackson, C. Grillet, et al., "Low propagation Loss Silicon-on-Sapphire Waveguides for the Mid-Infrared," *Optics Express* 19 (2011): 15212–15220.
85. M. A. Butt, "Emerging Trends in Thermo-Optic and Electro-Optic Materials for Tunable Photonic Devices," *Materials* 18 (2025): 2782.
86. R. S. E. Shamy, H. Mossad, and M. A. Swillam, "Silicon-on-Sapphire (SOS) Waveguide Modal Analysis for Mid-Infrared Applications," *Journal of Physics Communications* 1 (2017): 035011.
87. P. Muñoz, G. Micó, L. A. Bru, et al., "Silicon Nitride Photonic Integration Platforms for Visible, Near-Infrared and Mid-Infrared Applications," *Sensors* 17 (2017): 2088.

88. P. Muñoz, J. D. Doménech, C. Domínguez, et al, 2017 19th International Conference on Transparent Optical Networks (ICTON) (IEEE 2017), 1–4.
89. X. Guo, X. Ji, B. Yao, et al., “Ultra-Wideband Integrated Photonic Devices on Silicon Platform: From Visible to Mid-IR,” *Nanophotonics* 12 (2023): 167–196.
90. C. Xiang, W. Jin, and J. E. Bowers, “Silicon Nitride Passive and Active Photonic Integrated Circuits: Trends and Prospects,” *Photonics Research* 10 (2022): A82–A96.
91. Y. Lin, Z. Yong, X. Luo, et al., “Monolithically Integrated, Broadband, High-Efficiency Silicon Nitride-on-Silicon Waveguide Photodetectors in a Visible-Light Integrated Photonics Platform,” *Nature Communications* 13 (2022): 6362.
92. Q. Wilmart, H. El Dirani, N. Tyler, et al., “A Versatile Silicon-Silicon Nitride Photonics Platform for Enhanced Functionalities and Applications,” *Applied Sciences* 9 (2019): 255.
93. M. A. Butt, M. Słowikowski, D. Drecka, M. Jarosik, and R. Piramidowicz, “Experimental Characterization of a Silicon Nitride Asymmetric Loop-Terminated Mach-Zehnder Interferometer with a Refractive Index-Engineered Sensing Arm,” *Nanomaterials* 15 (2025): 1532.
94. L. Zhang, C. Cui, Y. Xue, P. Chen, and L. Fan, “Scalable Photonic-Phononic Integrated Circuitry for Reconfigurable Signal Processing,” *Nature Communications* 16 (2025): 2718.
95. A. Yi, C. Wang, L. Zhou, et al., “Silicon Carbide for Integrated Photonics,” *Applied Physics Reviews* 9 (2022): 031302.
96. H. Ou and X. Shi, “Novel Photonic Applications of Silicon Carbide,” *Materials* 16 (2023): 1014.
97. F. Garrisi, I. Chatzopoulos, R. Cernansky, and A. Politi, “Silicon Carbide Photonic Platform Based on Suspended Subwavelength Waveguides,” *Journal of the Optical Society of America B* 37 (2020): 3453–3460.
98. B. Lopez-Rodriguez, R. van der Kolk, S. Aggarwal, et al., “High-Quality Amorphous Silicon Carbide for Hybrid Photonic Integration Deposited at a Low Temperature,” *ACS Photonics* 10 (2023): 3748–3754.
99. F. Liu, H. Lv, H. Zhang, et al., “Design of polarization-Independent 1×2 Optical Power Splitter Based on Silicon Nitride Slot Waveguide Structure,” *Scientific Reports* 15 (2025): 25288.
100. L.-Y. Chen, Y.-J. Chen, D.-S. Wu, W.-H. Huang, and C.-T. Wang, “High-Sensitivity Silicon Nitride Optical Temperature Sensor Based on Cascaded Mach-Zehnder Interferometers,” *Optical Materials* 165 (2025): 117139.
101. P. Zhao, M. Karlsson, and P. A. Andrekson, *Optical Fiber Communication Conference (OFC) 2025 (2025) Paper M4D.3* (Optica Publishing Group, 2025), M4D.3.
102. D. Bose, M. W. Harrington, A. Isichenko, et al., “Anneal-Free Ultra-Low Loss Silicon Nitride Integrated Photonics,” *Light: Science & Applications* 13 (2024): 156.
103. M. Smit, X. Leijtens, H. Ambrosius, et al., “An Introduction to InP-Based Generic Integration Technology,” *Semiconductor Science and Technology* 29 (2014): 083001.
104. D. Kohler, G. Schindler, L. Hahn, et al., “Biophotonic Sensors with Integrated Si₃N₄-Organic Hybrid (SiNOH) Lasers for Point-of-Care Diagnostics,” *Light: Science & Applications* 10 (2021): 64.
105. A. Shams-Ansari, G. Huang, L. He, et al., “Reduced Material Loss in Thin-Film Lithium Niobate Waveguides,” *APL Photonics* 7 (2022): 081301.
106. Q. Zhao, R. O. Behunin, P. T. Rakich, et al., “Low-Loss Low Thermo-Optic Coefficient Ta₂O₅ on Crystal Quartz Planar Optical Waveguides,” *APL Photonics* 5 (2020): 116103.
107. S. Gundavarapu, G. M. Brodnik, M. Puckett, et al., “Sub-Hertz Fundamental Linewidth Photonic Integrated Brillouin Laser,” *Nature Photonics* 13 (2019): 60–67.
108. Y. Xie, J. Li, Y. Zhang, et al., “Soliton Frequency Comb Generation in CMOS-Compatible Silicon Nitride Microresonators,” *Photonics Research* 10 (2022): 1290–1296.
109. G. Xu, J. Yan, Z. Chen, et al., “Design of Germanium-Silicon Carbide Hybrid Waveguides for Mid-Infrared Third-Order Parametric Conversion,” *Optics Communications* 456 (2020): 124668.
110. U. Younis, S. K. Vanga, A. E.-J. Lim, P. G.-Q. Lo, A. A. Bettiol, and K.-W. Ang, “Germanium-on-SOI Waveguides for Mid-Infrared Wavelengths,” *Optics Express* 24 (2016): 11987–11993.
111. J. Lim, J. Shim, D.-M. Geum, and S. Kim, “Experimental Demonstration of Germanium-on-Silicon Slot Waveguides at Mid-Infrared Wavelength,” *IEEE Photonics Journal* 14 (2022): 5828709.
112. K. Gallacher, L. Baldassarre, R. W. Millar, et al., 2017 *IEEE 14th International Conference on Group IV Photonics (GFP)* (IEEE, 2017), 63–64.
113. Y.-C. Chang, V. Paeder, L. Hvozda, J.-M. Hartmann, and H. P. Herzig, “Low-Loss Germanium Strip Waveguides on Silicon For the Mid-Infrared,” *Optics Letters* 37 (2012): 2883–2885.
114. W. Li, P. Anantha, S. Bao, et al., “Germanium-on-Silicon Nitride Waveguides for Mid-Infrared Integrated Photonics,” *Applied Physics Letters* 109 (2016): 241101.
115. T.-H. Xiao, Z. Zhao, W. Zhou, et al., “High-Q Germanium Optical Nanocavity,” *Photonics Research* 6 (2018): 925–928.
116. P. Ma, D.-Y. Choi, Y. Yu, et al., “Low-Loss Chalcogenide Waveguides for Chemical Sensing in the Mid-Infrared,” *Optics Express* 21 (2013): 29927–29937.
117. R. Lin, F. Chen, X. Zhang, et al., “Mid-Infrared Optical Properties of Chalcogenide Glasses Within Tin-Antimony-Selenium Ternary System,” *Optics Express* 25 (2017): 25674–25688.
118. E. Zhu, Y. Liu, X. Sun, et al., “Correlation between Thermo-mechanical Properties and Network Structure in GeS_{100-x} Chalcogenide Glasses,” *Journal of Non-Crystalline Solids: X* 1 (2019): 100015.
119. R. Bystrický, S. K. Tiwari, P. Hutár, and M. Sýkora, “Thermal Stability of Chalcogenide Perovskites,” *Inorganic Chemistry* 63 (2024): 12826–12838.
120. J. Hu, V. Tarasov, N. Carlie, et al, *Ultrasensitive and Single-Molecule Detection Technologies II* (SPIE, 2007), 148–155.
121. J. Charrier, M.-L. Brandily, H. Lhermite, et al., “Evanescent Wave Optical Micro-Sensor Based on Chalcogenide Glass,” *Sensors and Actuators B: Chemical* 173 (2012): 468–476.
122. J. Hu, X. Sun, A. Agarwal, and L. C. Kimerling, “Design Guidelines for Optical Resonator Biochemical Sensors,” *Journal of the Optical Society of America B* 26 (2009): 1032–1041.
123. H. Lin, L. Li, Y. Zou, et al., “Demonstration of High-Q Mid-Infrared Chalcogenide Glass-on-Silicon Resonators,” *Optics Letters* 38 (2013): 1470–1472.
124. H. Lin, L. Li, F. Deng, et al., “Demonstration of Mid-Infrared Waveguide Photonic Crystal Cavities,” *Optics Letters* 38 (2013): 2779–2782.
125. A. Ganjoo, H. Jain, C. Yu, J. Irudayaraj, and C. G. Pantano, “Detection and Fingerprinting of Pathogens: Mid-IR Biosensor Using Amorphous Chalcogenide Films,” *Journal of Non-Crystalline Solids* 354 (2008): 2757–2762.
126. M. Pi, C. Zheng, H. Zhao, et al., “Mid-Infrared ChG-on-MgF₂ Waveguide Gas Sensor Based on Wavelength Modulation Spectroscopy,” *Optics Letters* 46 (2021): 4797–4800.
127. G. Granger, M. Bailly, H. Delahaye, et al, *Optica Advanced Photonics Congress 2022 Paper ATu3A.6* (Optica Publishing Group, 2022), ATu3A.6.
128. K. Zhang, G. Böhm, and M. A. Belkin, “Mid-Infrared Microring Resonators and Optical Waveguides on an InP Platform,” *Applied Physics Letters* 120 (2022): 061106.
129. H. Kalt, *Optical Properties of III–V Semiconductors* (Springer, 1996).

130. A. Hänsel and M. J. R. Heck, "Opportunities for Photonic Integrated Circuits in Optical Gas Sensors," *Journal of Physics: Photonics* 2 (2020): 012002.
131. K. M. Yoo, J. Midkiff, A. Rostamian, C. Chung, H. Dalir, and R. T. Chen, "InGaAs Membrane Waveguide: A Promising Platform for Monolithic Integrated Mid-Infrared Optical Gas Sensor," *ACS Sensors* 5 (2020): 861–869.
132. M. Razeghi, Q. Y. Lu, N. Bandyopadhyay, et al., "Quantum Cascade Lasers: From Tool to Product," *Optics Express* 23 (2015): 8462–8475.
133. P. Speier, U. Koerner, A. Nowitzki, F. Grotjahn, F. J. Tegude, and K. Wünnstel, "MOVPE studies for a monolithically integrated DH laser/HBT laser driver," *Journal of Crystal Growth* 93 (1988): 885–891.
134. J. B. D. Soole, H. Schumacher, H. P. LeBlanc, R. Bhat, and M. A. Koza, "High-Speed Performance of OMCVD Grown InAlAs/InGaAs MSM Photodetectors at 1.5 μm and 1.3 μm Wavelengths," *IEEE Photonics Technology Letters* 1 (1989): 250–252.
135. D. Cui, S. M. Hubbard, D. Pavlidis, A. Eisenbach, and C. Chelli, "Impact of Doping and MOCVD Conditions on Minority Carrier Lifetime of Zinc- and Carbon-Doped InGaAs and Its Applications to Zinc- and Carbon-Doped InP/InGaAs Heterostructure Bipolar Transistors," *Semiconductor Science and Technology* 17 (2002): 503.
136. T. S. Karnik, K. P. Dao, Q. Du, et al., "High-Efficiency Mid-Infrared InGaAs/InP Arrayed Waveguide Gratings," *Optics Express* 31 (2023): 5056–5068.
137. K. L. Vodopyanov, I. Makasyuk, and P. G. Schunemann, "Grating Tunable 4 - 14 μm GaAs Optical Parametric Oscillator Pumped at 3 μm ," *Optics Express* 22 (2014): 4131–4136.
138. F. Gutty, A. Grisard, C. Larat, et al., "High Peak-Power Laser System Tuneable From 8 to 10 μm ," *Advanced Optical Technologies* 6 (2017): 95–101.
139. S. Roux, L. Cerutti, E. Tournie, et al., "Low-Loss Orientation-Patterned GaSb Waveguides for Mid-Infrared Parametric Conversion," *Optical Materials Express* 7 (2017): 3011–3016.
140. J. Haas, P. Artmann, and B. Mizaikoff, "Mid-Infrared GaAs/AlGaAs Micro-Ring Resonators Characterized via Thermal Tuning," *RSC Advances* 9 (2019): 8594–8599.
141. Q. Ma, G. Ren, A. Mitchell, and J. Z. Ou, "Recent Advances on Hybrid Integration of 2D Materials on Integrated Optics Platforms," *Nanophotonics* 9 (2020): 2191–2214.
142. W. Li, H. Li, K. Khan, et al., "Infrared Light Emission Devices Based on Two-Dimensional Materials," *Nanomaterials* 12 (2022): 2996.
143. A. K. Geim and K. S. Novoselov, "The Rise of Graphene," *Nature Materials* 6 (2007): 183–191.
144. J. Wu, H. Ma, P. Yin, et al., "Two-Dimensional Materials for Integrated Photonics: Recent Advances and Future Challenges," *Small Science* 1 (2021): 2000053.
145. K. S. Novoselov, A. K. Geim, S. V. Morozov, et al., "Two-Dimensional Gas of Massless Dirac Fermions in Graphene," *Nature* 438 (2005): 197–200.
146. E. A. Anyebe, I. Sandall, Z. M. Jin, et al., "Optimization of Self-Catalyzed InAs Nanowires on Flexible Graphite for Photovoltaic Infrared Photodetectors," *Scientific Reports* 7 (2017): 46110.
147. E. Kahn, "Functional Hetero-Interfaces in Atomically Thin Materials," *Materials Today* 37 (2020): 74–92.
148. Y. Fang, Y. Ge, C. Wang, and H. Zhang, "Mid-Infrared Photonics Using 2D Materials: Status and Challenges," *Laser & Photonics Reviews* 14 (2020): 1900098.
149. C.-C. Huang, R.-J. Chang, and C.-W. Cheng, "Ultra-Low-Loss Mid-Infrared Plasmonic Waveguides Based on Multilayer Graphene Metamaterials," *Nanomaterials* 11 (2021): 2981.
150. H. Li and Z. Yang, "Recent Progress in Mid-Infrared Photodetection Devices Using 2D/nD (n=0, 1, 2, 3) Heterostructures," *Materials & Design* 225 (2023): 111446.
151. N. Negm, S. Zayouna, S. Parhizkar, et al., "Graphene Thermal Infrared Emitters Integrated Into Silicon Photonic Waveguides," *ACS Photonics* 11 (2024): 2961–2969.
152. Y. Gao, W. Zhou, X. Sun, H. K. Tsang, and C. Shu, "Cavity-Enhanced Thermo-Optic Bistability and Hysteresis in a Graphene-on-Si₃N₄ Ring Resonator," *Optics Letters* 42 (2017): 1950–1953.
153. H. Pan, H. Chu, Z. Pan, et al., "Large-Scale Monolayer Molybdenum Disulfide (MoS₂) for Mid-Infrared Photonics," *Nanophotonics* 9 (2020): 4703–4710.
154. A. J. Sternbach, S. Latini, S. Chae, et al., "Femtosecond Exciton Dynamics in WSe₂ Optical Waveguides," *Nature Communications* 11 (2020): 3567.
155. C. Liu, J. Guo, L. Yu, et al., "Silicon/2D-material Photodetectors: From Near-Infrared to Mid-Infrared," *Light: Science & Applications* 10 (2021): 123.
156. S. Edelstein, S. R. K. C. Indukuri, N. Mazurski, and U. Levy, "Waveguide-Integrated Mid-IR Photodetector and All-Optical Modulator Based on Interlayer Excitons Absorption in a WS₂/hfs 2 Heterostructure," *Nanophotonics* 11 (2022): 4337–4345.
157. S. Parhizkar, M. Prechtel, A. L. Giesecke, et al., "Two-Dimensional Platinum Diselenide Waveguide-Integrated Infrared Photodetectors," *ACS Photonics* 9 (2022): 859–867.
158. X. Zhang, Y. Cheng, and Y. Zhang, "Mid-Infrared Ridge Waveguides Fabricated in CaF₂ Crystal by the Technique of O⁵⁺ Ion Irradiation," *Applied Optics* 60 (2021): 6302–6307.
159. L. Zheng, J. Zhao, Y. Wang, et al., "Mid-IR Optical Property of Dy:CaF₂-SrF₂ Crystal Fabricated by Multicrucible Temperature Gradient Technology," *Crystals* 11 (2021): 907.
160. L. Li, W. Nie, Z. Li, et al., "All-Laser-Micromachining of Ridge Waveguides in LiNbO₃ Crystal for Mid-Infrared Band Applications," *Scientific Reports* 7 (2017): 7034.
161. R. He, Q. An, J. R. V. de Aldana, Q. Lu, and F. Chen, "Femtosecond-Laser Micromachined Optical Waveguides in Bi₄Ge₃O₁₂ Crystals," *Applied Optics* 52 (2013): 3713–3718.
162. D. A. Biasetti, E. J. Di Liscia, and G. A. Torchia, "Optical Waveguides Fabricated in Cr:LiSAF by Femtosecond Laser Micromachining," *Optical Materials* 73 (2017): 25–32.
163. M. A. Butt, H.-D. Nguyen, A. Ródenas, et al., "Low-Repetition Rate Femtosecond Laser Writing of Optical Waveguides in KTP Crystals: Analysis of Anisotropic Refractive Index Changes," *Optics Express* 23 (2015): 15343–15355.
164. G. Corrielli, A. Crespi, and R. Osellame, "Femtosecond Laser Micromachining for Integrated Quantum Photonics," *Nanophotonics* 10 (2021): 3789–3812.
165. M. A. Butt and R. Piramidowicz, "Suspended Slot Membrane Waveguide Based on Germanium-on-Silicon-on-Insulator at $\lambda = 4.23 \mu\text{m}$ for CO₂ Monitoring," *Micromachines* 15 (2024): 1434.
166. M. A. Butt and R. Piramidowicz, "Integrated Photonic Sensors for the Detection of Toxic Gases—A Review," *Chemosensors* 12 (2024): 143.
167. M. A. Butt, "Dielectric Waveguide-Based Sensors with Enhanced Evanescent Field: Unveiling the Dynamic Interaction with the Ambient Medium for Biosensing and Gas-Sensing Applications—A Review," *Photonics* 11 (2024): 198.
168. N. L. Kazanskiy, S. N. Khonina, and M. A. Butt, "Polarization-Insensitive Hybrid Plasmonic Waveguide Design for Evanescent Field Absorption Gas Sensor," *Photonic Sensors* 11 (2021): 279–290.
169. S. N. Khonina, N. L. Kazanskiy, and M. A. Butt, "Evanescent Field Ratio Enhancement of a Modified Ridge Waveguide Structure for Methane Gas Sensing Application," *IEEE Sensors Journal* 20 (2020): 8469–8476.
170. M. A. Butt and R. Piramidowicz, "Standard Slot Waveguide and Double Hybrid Plasmonic Waveguide Configurations for Enhanced

- Evanescent Field Absorption methane Gas Sensing,” *Photonics Letters of Poland* 14 (2022): 10–12.
171. M. A. Butt, S. A. Degtyarev, S. N. Khonina, and N. L. Kazanskiy, “An Evanescent Field Absorption Gas Sensor at Mid-IR 3.39 μm Wavelength,” *Journal of Modern Optics* 64 (2017): 1892–1897.
 172. Y. Huang, S. K. Kalyoncu, Q. Song, and O. Boyraz, paper presented at 2012 Conference on Lasers and Electro-Optics (CLEO), San Jose, California, USA, May 6–11 2012.
 173. M. A. Butt, S. N. Khonina, and N. L. Kazanskiy, “Silicon on Silicon Dioxide Slot Waveguide Evanescent Field Gas Absorption Sensor,” *Journal of Modern Optics* 65 (2018): 174–178.
 174. K. Joshi, B. L. Meena, and K. Gehlot, “Performance of Silicon-on-Insulator and Silicon-on-Sapphire Based Evanescent Field Gas Sensor Operating at 2.86 μm ,” *AIP Conference Proceedings* 2768 (2023): 020023.
 175. C. Ranacher, C. Consani, N. Vollert, et al., “Characterization of Evanescent Field Gas Sensor Structures Based on Silicon Photonics,” *IEEE Photonics Journal* 10 (2018): 2700614.
 176. F. K. Tittel, R. Lewicki, M. Jahjah, et al, *Terahertz and Mid Infrared Radiation: Detection of Explosives and CBRN (Using Terahertz)*, eds. M. F. Pereira, O. Shulika (Springer Netherlands, 2014), 153–165.
 177. M. A. Butt, G. S. Voronkov, E. P. Grakhova, R. V. Kutluyarov, N. L. Kazanskiy, and S. N. Khonina, “Environmental Monitoring: A Comprehensive Review on Optical Waveguide and Fiber-Based Sensors,” *Biosensors* 12 (2022): 1038.
 178. A. Baldacci, V. Malathy Devi, D.-W. Chen, K. Narahari Rao, and B. Fridovich, “Absorption Spectrum of Carbon Dioxide at 4.3 μm ,” *Journal of Molecular Spectroscopy* 70 (1978): 143–159.
 179. M. A. Butt, “Analyzing the Evanescent Field Ratio of Ridge Waveguide Based on Different Material Platforms for Sensing Applications,” *Journal of Optics* 26 (2024): 095803.
 180. C. Ranacher, A. Tortschanoff, C. Consani, M. Moridi, T. Grille, and B. Jakoby, “Photonic Gas Sensor Using a Silicon Strip Waveguide,” *Proceedings* 1 (2017): 547.
 181. I. K. Ilev and R. W. Waynant, in *Mid-Infrared Semiconductor Optoelectronics*, ed. A. Krier (Springer, 2006), 615–634.
 182. K. Isensee, N. Kröger-Lui, and W. Petrich, “Biomedical Applications of Mid-Infrared Quantum Cascade Lasers – A Review,” *The Analyst* 143 (2018): 5888–5911.
 183. H. M. Heise, L. Küpper, and L. N. Butvina, “Bio-Analytical Applications of Mid-Infrared Spectroscopy Using Silver Halide Fiber-Optic Probes,” *Spectrochimica Acta Part B: Atomic Spectroscopy* 57 (2002): 1649–1663.
 184. C. Hughes and M. J. Baker, “Can Mid-Infrared Biomedical Spectroscopy of Cells, Fluids and Tissue Aid Improvements in Cancer Survival? A Patient Paradigm,” *The Analyst* 141 (2016): 467–475.
 185. R. Kasahara, S. Kino, S. Soyama, and Y. Matsuura, “Noninvasive Glucose Monitoring Using Mid-Infrared Absorption Spectroscopy Based on a Few Wavenumbers,” *Biomedical Optics Express* 9 (2018): 289–302.
 186. R. Vitorino, A. S. Barros, S. Guedes, D. C. Caixeta, and R. Sabino-Silva, “Diagnostic and Monitoring Applications Using Near Infrared (NIR) Spectroscopy in Cancer and Other Diseases,” *Photodiagnosis and Photodynamic Therapy* 42 (2023): 103633.
 187. G. F. Rangel, L. Diaz de León Martínez, L. S. Walter, and B. Mizaikoff, “Recent Advances and Trends in Mid-Infrared Chem/Bio Sensors,” *TrAC Trends in Analytical Chemistry* 180 (2024): 117916.
 188. K. J. Linden, et al., San Francisco, California, United States (2013).
 189. A. Joshi, D. B. Bennett, and O. M. Stafsudd, “Monitoring Corneal Hydration with a Mid-Infrared (IR) Laser,” *The Ocular Surface* 13 (2015): 43–46.
 190. K. Zhu, J. Shen, W. Xu, et al., “Application of Serum Mid-Infrared Spectroscopy Combined with Machine Learning in Rapid Screening of Breast Cancer and Lung Cancer,” *International Journal of Intelligent Systems* 2023 (2023): 4533108.
 191. E. Kontsek, A. Pesti, M. Björnstedt, et al., “Mid-Infrared Imaging is Able to Characterize and Separate Cancer Cell Lines,” *Pathology & Oncology Research* 26 (2020): 2401–2407.
 192. B. Guo, Y. Wang, C. Peng, et al., “Laser-Based Mid-Infrared Reflectance Imaging of Biological Tissues,” *Optics Express* 12 (2004): 208–219.
 193. S. Wartewig and R. H. H. Neubert, “Pharmaceutical Applications of Mid-IR and Raman Spectroscopy,” *Advanced Drug Delivery Reviews* 57 (2005): 1144–1170.
 194. Y. Meng, Y. Chen, L. Lu, et al., “Optical Meta-Waveguides for Integrated Photonics and Beyond,” *Light: Science & Applications* 10 (2021): 235.
 195. Y. Zou, S. Chakravarty, C.-J. Chung, X. Xu, and R. T. Chen, “Mid-Infrared Silicon Photonic Waveguides and Devices [Invited],” *Photonics Research* 6 (2018): 254–276.
 196. T. Xu, Y. Dong, Q. Zhong, et al., “Mid-Infrared Integrated Electro-Optic Modulators: A Review,” *Nanophotonics* 12 (2023): 3683–3706.
 197. A. Yadav and A. Agarwal Murthy, “Integrated Photonic Materials for the Mid-Infrared,” *International Journal of Applied Glass Science* 11 (2020): 491–510.
 198. N. S. Prasad, in *Free-Space Laser Communications: Principles and Advances*, eds. A. K. Majumdar, J. C. Ricklin (Springer, 2008), 347–391.
 199. Y. Yao, A. J. Hoffman, and C. F. Gmachl, “Mid-Infrared Quantum Cascade Lasers,” *Nature Photonics* 6 (2012): 432–439.
 200. P. Didier, H. Knötig, O. Spitz, et al., “Interband Cascade Technology for Energy-Efficient Mid-Infrared Free-Space Communication,” *Photonics Research* 11 (2023): 582–590.
 201. G. Marschick, J. Pelini, T. Gabbrielli, et al., “Mid-infrared Ring Interband Cascade Laser: Operation at the Standard Quantum Limit,” *ACS Photonics* 11 (2024): 395–403.
 202. Q. Hao, G. Zhu, S. Yang, et al., “Mid-Infrared Transmitter and Receiver Modules for Free-Space Optical Communication,” *Applied Optics* 56 (2017): 2260–2264.
 203. J. Chiles, N. Nader, E. J. Stanton, et al., “Multifunctional Integrated Photonics in the Mid-Infrared with Suspended AlGaAs on Silicon,” *Optica* 6 (2019): 1246–1254.
 204. H. Gao, W. Yan, D. Zhang, and D. Yu, “Heterogeneous Integration Technology Drives the Evolution of Co-Packaged Optics,” *Micromachines* 16 (2025): 1037.
 205. J. Chiles and S. Fathpour, “Mid-Infrared Integrated Waveguide Modulators Based on Silicon-on-Lithium-Niobate Photonics,” *Optica* 1 (2014): 350–355.
 206. J. Y. Tan, P. J. Ker, K. Y. Lau, M. A. Hannan, and S. G. H. Tang, “Applications of Photonics in Agriculture Sector: A Review,” *Molecules* 24 (2019): 2025.
 207. D. Nath, R. Laik, V. S. Meena, et al., “Strategies to Admittance Soil Quality Using Mid-Infrared (mid-IR) Spectroscopy an Alternate Tool for Conventional Lab Analysis: A Global Perspective,” *Environmental Challenges* 7 (2022): 100469.
 208. D. L. Elmore, C. A. Lendon, S. A. Smith, and C. L. Leverette, *Spectrochemical Analysis Using Infrared Multichannel Detectors* (Wiley, 2007).
 209. E. Mendes and N. Duarte, “Mid-Infrared Spectroscopy as a Valuable Tool to Tackle Food Analysis: A Literature Review on Coffee, Dairies, Honey, Olive Oil and Wine,” *Foods* 10 (2021): 477.
 210. E. Hatzakis, “Nuclear Magnetic Resonance (NMR) Spectroscopy in Food Science: A Comprehensive Review,” *Comprehensive Reviews in Food Science and Food Safety* 18 (2019): 189–220.

211. S. N. Khonina, N. L. Kazanskiy, I. V. Oseledets, A. V. Nikonov, and M. A. Butt, "Synergy Between Artificial Intelligence and Hyperspectral Imaging—A Review," *Technologies* 12 (2024): 163.
212. M. Kucharska and J. Grabka, "A Review of Chromatographic Methods for Determination of Synthetic Food Dyes," *Talanta* 80 (2010): 1045–1051.
213. R. H. Wilson and H. S. Tapp, "Mid-Infrared Spectroscopy for Food Analysis: Recent New Applications and Relevant Developments in Sample Presentation Methods," *TrAC Trends in Analytical Chemistry* 18 (1999): 85–93.
214. W.-H. Su and D.-W. Sun, "Mid-Infrared (MIR) Spectroscopy for Quality Analysis of Liquid Foods," *Food Engineering Reviews* 11 (2019): 142–158.
215. B. H. Stuart, *Infrared Spectroscopy: Fundamentals and Applications* (John Wiley & Sons, 2004).
216. O. Q. Adiamo, Y. Sultanbawa, and D. Cozzolino, "Mid-Infrared Spectroscopy as a Rapid Tool to Qualitatively Predict the Effects of Species, Regions and Roasting on the Nutritional Composition of Australian Acacia Seed Species," *Molecules* 26 (2021): 1879.
217. M. Malinauskas, A. Žukauskas, S. Hasegawa, et al., "Ultrafast Laser Processing of Materials: From Science to Industry," *Light: Science & Applications* 5 (2016): 16133.
218. D. Schuocker, "LASER CUTTING," *Materials and Manufacturing Processes* 4 (1989): 311–330.
219. S. Genna, E. Menna, G. Rubino, and V. Tagliaferri, "Experimental Investigation of Industrial Laser Cutting: The Effect of the Material Selection and the Process Parameters on the Kerf Quality," *Applied Sciences* 10 (2020): 4956.
220. A. C. Tam, *Advanced Solid State Lasers, Paper PT2* (Optica Publishing Group, 1996), PT2.
221. D. Kazakov, T. P. Letsou, M. Piccardo, et al., "Driven Bright Solitons on a Mid-Infrared Laser Chip," *Nature* 641 (2025): 83–89.
222. R. T. Chen, "Polymer-Based Photonic Integrated Circuits," *Optics & Laser Technology* 25 (1993): 347–365.
223. S. F. Iftekar, A. Aabid, A. Amir, and M. Baig, "Advancements and Limitations in 3D Printing Materials and Technologies: A Critical Review," *Polymers* 15 (2023): 2519.
224. M. A. Butt, "Thin-Film Coating Methods: A Successful Marriage of High-Quality and Cost-Effectiveness—A Brief Exploration," *Coatings* 12 (2022): 1115.
225. M. A. Butt, "Integrated Optics: Platforms and Fabrication Methods," *Encyclopedia* 3 (2023): 824–838.
226. K. Yasuhara, F. Yu, and T. Ishigure, "Circular core Single-Mode Polymer Optical Waveguide Fabricated Using the Mosquito Method with Low Loss at 1310/1550 nm," *Optics Express* 25 (2017): 8524–8533.
227. T. Merkle, R. Götz, J.-Y. Choi, and S. Koch, "Polymer Multichip Module Process Using 3-D Printing Technologies for D-Band Applications," *IEEE Transactions on Microwave Theory and Techniques* 63 (2015): 481–493.
228. W. M. Pätzold, C. Reinhardt, A. Demircan, and U. Morgner, "Cascaded-Focus Laser Writing of Low-Loss Waveguides in Polymers," *Optics Letters* 41 (2016): 1269–1272.
229. C. Frayssinous, V. Fortin, J.-P. Bérubé, A. Fraser, and R. Vallée, "Resonant Polymer Ablation Using a Compact 3.44 μm Fiber Laser," *Journal of Materials Processing Technology* 252 (2018): 813–820.
230. J.-P. Bérubé, C. Frayssinous, J. Lapointe, S. Duval, V. Fortin, and R. Vallée, "Direct Inscription of On-Surface Waveguides in Polymers Using a Mid-ir Fiber Laser," *Optics Express* 27 (2019): 31013–31022.
231. K.-K. Choi, A. K. Dutta, and N. K. Dhar, "High Performance Infrared Photodetectors and Energy Harvesting Devices Based on Micro-Scaled Photonics Structures," *IEEE Journal of Quantum Electronics* 57 (2021): 4000310.
232. S. J. Byrnes, R. Blanchard, and F. Capasso, "Harvesting Renewable Energy From Earth's Mid-Infrared Emissions," *Proceedings of the National Academy of Sciences* 111 (2014): 3927–3932.
233. A. Georgiadis, G. Vera Andia, and A. Collado, "Rectenna Design and Optimization Using Reciprocity Theory and Harmonic Balance Analysis for Electromagnetic (EM) Energy Harvesting," *IEEE Antennas and Wireless Propagation Letters* 9 (2010): 444–446.
234. S. V. Boriskina, M. A. Green, K. Catchpole, et al., "Roadmap on Optical Energy Conversion," *Journal of Optics* 18 (2016): 073004.
235. Y. Nishijima, R. Komatsu, T. Yamamura, A. Balčytis, G. Seniutinas, and S. Juodkazis, "Design Concept of a Hybrid Photo-Voltaic/Thermal Conversion Cell for Mid-Infrared Light Energy Harvester," *Optical Materials Express* 7 (2017): 3484–3493.
236. L. Huang, B. Dong, Z. G. Yu, et al., "Mid-Infrared Modulators Integrating Silicon and Black Phosphorus Photonics," *Materials Today Advances* 12 (2021): 100170.
237. S. C. Chaparala, F. Xie, C. Caneau, L. C. Hughes, and C. Zah, "Thermal Management of Mid-Infrared (IR) Quantum Cascade Lasers," in *2010 Proceedings 60th Electronic Components and Technology Conference (ECTC)*, (IEEE 2010), 693–699.
238. S. C. Chaparala, F. Xie, C. Caneau, C. E. Zah, and L. C. Hughes, "Design Guidelines for Efficient Thermal Management of Mid-Infrared Quantum Cascade Lasers," *IEEE Transactions on Components, Packaging and Manufacturing Technology* 1 (2011): 1975–1982.
239. Y. Nishijima, S. Kimura, Y. Takeshima, and S. Juodkazis, "Metasurfaces as Energy Valves for Sustainable Energy Management," *Micromachines* 13 (2022): 1769.
240. O. Benderov, I. Nepochurenko, B. Stepanov, et al., "Broadband mid-IR Chalcogenide Fiber Couplers," *Applied Optics* 58 (2019): 7222–7226.
241. W. Tian, Y. Wang, H. Dang, H. Hou, and Y. Xi, "Photonic Integrated Circuits: Research Advances and Challenges in Interconnection and Packaging Technologies," *Photonics* 12 (2025): 821.
242. M. S. Vitiello, G. Scalari, B. Williams, and P. D. Natale, "Quantum Cascade Lasers: 20 Years of Challenges," *Optics Express* 23 (2015): 5167–5182.
243. M. Fagot, D. A. Díaz-Thomas, A. Gilbert, et al., "Interband Cascade Lasers Grown Simultaneously on GaSb, GaAs and Si Substrates," *Optics Express* 32 (2024): 11057–11064.
244. P. Ashok and M. Ganesh Madhan, "Impedance Characteristics of Mid Infrared Quantum Cascade Lasers," *Optics & Laser Technology* 134 (2021): 106662.
245. O. Spitz, J. Wu, A. Herdt, et al., "Extreme Events in Quantum Cascade Lasers," *Advanced Photonics* 2 (2020): 066001.
246. Y. Bidaux, A. Bismuto, P. Patimisco, et al., "Mid Infrared Quantum Cascade Laser Operating in Pure Amplitude Modulation for Background-Free Trace Gas Spectroscopy," *Optics Express* 24 (2016): 26464–26471.
247. Y.-F. C. Chau, "Nanophotonic Materials and Devices: Recent Advances and Emerging Applications," *Micromachines* 16 (2025): 933.
248. J. Zhou, X. Liu, and H. Zhou, "Artificial-Intelligence-Enhanced Mid-Infrared Lab-on-a-Chip for Mixture Spectroscopy Analysis," *Laser & Photonics Reviews* 19 (2025): 2400754.
249. Y. Nishijima, N. To, A. Balčytis, and S. Juodkazis, "Absorption and Scattering in Perfect Thermal Radiation Absorber-Emitter Metasurfaces," *Optics Express* 30 (2022): 4058–4070.
250. N. To, S. Juodkazis, and Y. Nishijima, "Detailed Experiment-Theory Comparison of Mid-Infrared Metasurface Perfect Absorbers," *Micromachines* 11 (2020): 409.

Biographies



Muhammad A. Butt received his Ph.D. in Materials Science from Universitat Rovira i Virgili, Spain, in 2015. He is currently serving as an Assistant Professor at the Warsaw University of Technology, Poland. In 2018, he held the position of Research Assistant Professor at Nicolaus Copernicus University, Poland. His international research experience includes a stay at the Optoelectronics Research Centre (ORC), University of Southampton, United Kingdom, in 2013, and a recent research visit to Universitat Politècnica

de València (UPV), Spain, in 2024. Dr. Butt has authored over 200 peer-reviewed scientific publications and book chapters. His research contributions have been recognized globally, and he was listed among the World's Top 2% Most Cited Researchers in 2021-2025. He also serves as an Editorial Board Member of Scientific Reports. His research interests encompass integrated photonics, metamaterials, numerical modelling, light-matter interaction, and plasmonic sensors.



Daniel Pastor Abellán received the Ph.D. degree in Electrical Engineering and Telecommunications from the Universitat Politècnica de València (UPV), Spain, in 1996, with a dissertation on diffraction gratings in optical fibers for photonic signal processing. He is currently a Full Professor with the Photonic Research Lab (PRL) at UPV. His research interests include photonic integrated circuits, hybrid integration technologies, optical filters, LiDAR systems, and photonic sensing technologies in the

NIR and MIR spectral bands. He has authored or coauthored more than 200 scientific papers and has been involved in multiple national and European research projects. He also serves as Chief Scientific Officer at HYBPIC, a UPV spin-off focused on hybrid photonic components.



Ryszard Piramidowicz PhD, DSc, MBA. Associate Professor and Head of the Optoelectronics Division at the Institute of Microelectronics and Optoelectronics, Warsaw University of Technology (WUT). Director of the Photonic Integrated Circuits Development Program at VIGO Photonics. Co-founder and CEO of Light-House - the first Polish integrated photonics design house, a WUT spin-off. A recognized expert in optical spectroscopy, solid-state laser technologies, fiber-optic communications, and integrated photonics.

Author of over 400 peer-reviewed journal and conference publications. Principal investigator, coordinator, or leader of more than 30 R&D and investment projects in optoelectronics. President of the Polish Committee of Optoelectronics of the Association of Polish Electrical Engineers (SEP). Member of IEEE, Optica, and SPIE. Member of the Photonics21 Board of Stakeholders.